

Review from 2F

1-D Schrodinger eq.

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x,t)}{\partial x^2} + V(x) \Psi(x,t) = i\hbar \frac{\partial}{\partial t} \Psi(x,t)$$

Time-independent 1-D Schrodinger eq.

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x) \psi(x) = E \psi(x)$$

Operators

$$\hat{E} = i\hbar \frac{\partial}{\partial t}, \quad \hat{p} = -i\hbar \frac{\partial}{\partial x}, \quad \hat{x} = x$$

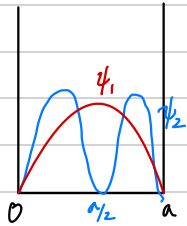
Probability

$$|\Psi(x,t)|^2 = P(x,t)$$

Expected value

$$\text{ex } \langle \hat{p} \rangle = -i\hbar \langle \psi | \frac{\partial}{\partial x} | \psi \rangle = -i\hbar \int_{-\infty}^{\infty} \psi^*(x) \frac{\partial}{\partial x} \psi(x) dx$$

Infinite sq. well



$$\text{Energy, } E_n = \frac{\pi^2 \hbar^2 n^2}{2ma^2} = \frac{\hbar^2 k^2}{2m} \quad \text{+0}$$

$$\text{Eigenstates: } \psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

$$\text{Momentum, } p_n = \pm \hbar k = \pm \frac{\hbar n \pi}{a} \quad (\text{left/right})$$

$$k = n\pi/a$$

Uncertainty Principle

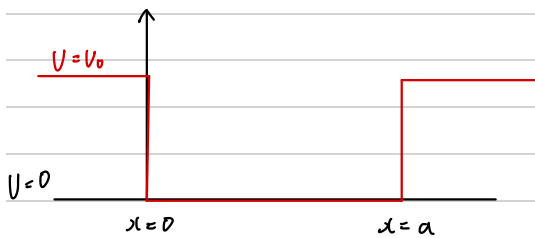
$$\Delta x \Delta p \geq \hbar/2 \sim O(\hbar)$$

$$\Delta E \Delta t \geq \hbar/2 \sim O(\hbar)$$

Orthonormality

$$\int \psi_n^* \psi_m dx = \delta_{nm} = \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n \end{cases}$$

7.8 Non-rigid box (Non infinite well)



$$V(x) = \begin{cases} 0, & 0 \leq x \leq a \\ V_0, & x < 0; x > a \end{cases}$$

$$\alpha = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

$$k = \sqrt{\frac{2m \cdot E}{\hbar^2}}$$

Rules

ψ, ψ' cont. on transition points

ψ'' changes sign on transition points

$$E_{\text{infinite}} > E_{\text{finite}} \quad \text{as } E \propto 1/a^2$$

For $x < 0$,

$$\psi(x) = A_0 e^{+\alpha x}$$

For $x > a$,

$$\psi(x) = A_1 e^{-\alpha x}$$

For $0 \leq x \leq a$,

$$\psi(x) = B e^{ikx} + C e^{-ikx}$$

(oscillations)

7.9 Simple Harmonic Oscillator Potential

↳ For $V(x) = \frac{1}{2} kx^2$

$$\text{Must satisfy } -\frac{\hbar^2}{2m} \psi'' + \frac{1}{2} kx^2 \psi = E_n \psi$$

$$\psi_n(x) = P_n(x) e^{-x^2/2b^2}, \quad b^2 = \frac{\hbar}{m\omega_c} \quad [m^2], \quad \omega_c = \sqrt{\frac{k}{m}}$$

↳ P_n are Hermite Polynomials

$$\begin{cases} P_0(x) = A_0 \\ P_1(x) = A_1 x/b \\ P_2(x) = A_2 \left(1 - \frac{2x^2}{b^2}\right) \\ \vdots \end{cases}, \quad A_0 = \frac{1}{(\pi b^2)^{1/4}}$$

↳ Approximation of potential $V(x) = \frac{1}{2} kx^2$ [Parabolic potential & Linear force]

↳ ex. Diatomic molecules H_2, HCl, HCR

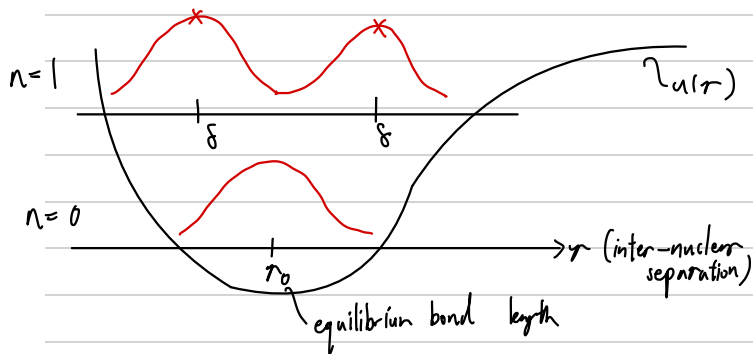
$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega_c, \quad n = 0, 1, 2, \dots$$

↳ $V(x) \rightarrow \infty$ as $x \rightarrow \infty$

∞ # of bound states

↳ $|\psi_n(x)|^2 \rightarrow$ Probability distribution for E_n in $\frac{1}{2}kx^2$ potential

$|\psi(r_0)|^2 \rightarrow$ Probability distribution for separation of atoms in a diatomic molecule (at particular energy level)



If $E = E_1 = \frac{3}{2} \hbar \omega_c$
expect to find at $r_0 \pm \delta$

7.1 Time dependent Schrodinger eqn.

$$\hookrightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x,t)}{\partial x^2} + V(x) \Psi(x,t) = i\hbar \frac{\partial}{\partial t} \Psi(x,t)$$

↳ Use separation of variables

$$\Psi(x,t) = \underbrace{\psi(x)}_{\text{space}} \underbrace{e^{i\omega t}}_{\text{time}}$$

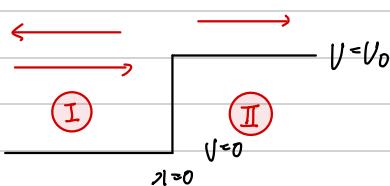
Combination indep. solns of infinite sq. well

$$\begin{aligned} \psi(x,t) &= a e^{i\omega_1 t} \psi_1(x) + b e^{i\omega_2 t} \psi_2(x) \\ &= e^{i\omega_1 t} [a \psi_1(x) + b e^{i\omega_{21} t} \psi_2(x)] \quad \omega_{21} = \omega_2 - \omega_1 \end{aligned}$$

$$\begin{aligned} |\psi(x,t)|^2 &= |a \psi_1(x) + b e^{i\omega_{21} t} \psi_2(x)|^2 \\ &= |a \psi_1(x) + \psi_2(x) [\cos \omega_{21} t + i \sin \omega_{21} t]|^2 \\ &= [a \psi_1(x) + b \psi_2(x) \cos \omega_{21} t]^2 + \psi_2^2(x) \sin^2 \omega_{21} t \\ &= a^2 \psi_1^2(x) + b^2 \psi_2^2(x) + \underbrace{2ab \psi_1(x) \psi_2(x) \cos \omega_{21} t}_{\text{time dependence}} \end{aligned}$$

↳ Moment in QM happens when energy levels interfere

7.10 Scattering, Transmission at finite step



↳ For particles $E > V_0$,

$$\begin{aligned} \psi_I(x) &= A_0 e^{ik_1 x} + A e^{-ik_1 x} \\ \psi_{II}(x) &= B e^{ik_2 x} \end{aligned} \quad \begin{aligned} k_1 &= \sqrt{\frac{2mE}{\hbar^2}} \\ k_2 &= \sqrt{\frac{2m(E-V_0)}{\hbar^2}} \end{aligned}$$

↳ Reflection prob, $R = \frac{|A|^2}{|A_0|^2} = \frac{(k_1 - k_2)^2}{(k_1 + k_2)^2}$

↳ Transmission prob, $T = 1 - R = \frac{k_2}{k_1} \frac{|B|^2}{|A_0|^2} = \frac{4k_1 k_2}{(k_1 + k_2)^2}$

↳ Boundary cond.

$$\begin{cases} \psi_I(0) = \psi_{II}(0) \rightarrow A_0 + A = B \\ \psi_I'(0) = \psi_{II}'(0) \rightarrow k_1(A_0 - A) = k_2 B \end{cases}$$

$$A = \frac{k_1 - k_2}{k_1 + k_2} A_0, \quad B = \frac{2k_1}{k_1 + k_2} A_0$$

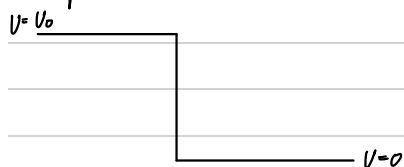
↳ Classically, $R=0, T=1$

↳ QM, $R>0, T<1$

$E \gg V_0, k_1 \sim k_2$

↳ If $E < V_0, T=0, R=1$

Step down (same)



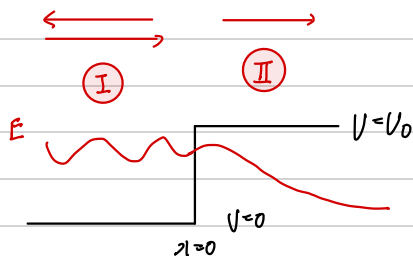
Systems described: Electron gun

As electrons cross $x=0$,

$$E \rightarrow E - V_0 \rightarrow \text{Slower movement} \rightarrow V_2/V_1 = k_2/k_1$$

Represents change in particle speed

$E < V_0$



$$\begin{aligned} \psi_I(x) &= A_0 e^{ik_1 x} + A e^{-ik_1 x} \\ \psi_{II}(x) &= B e^{-\alpha x} \end{aligned}$$

$$R = \frac{|A|^2}{|A_0|^2} = \left(\frac{ik_1 + \alpha}{ik_1 - \alpha} \right) \left(\frac{-ik_1 + \alpha}{-ik_1 - \alpha} \right) = 1$$

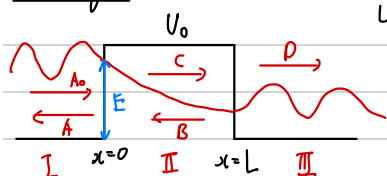
Bound and.

$T=0$

$$\begin{cases} A_0 + A = B \\ ik_1 A_0 - ik_1 A = -\alpha B \end{cases}$$

$$A = \frac{ik_1 + \alpha}{ik_1 - \alpha} A_0, \quad B = \frac{2ik_1}{ik_1 - \alpha} A_0$$

Tunneling



$$\begin{cases} \psi_I(x) = A_0 e^{ikx} + A e^{-ikx} \\ \psi_{II}(x) = B e^{-\alpha x} + C e^{\alpha x} \\ \psi_{III}(x) = D e^{ikx} \end{cases}, \quad k = \frac{\sqrt{2mE}}{\hbar}, \quad \alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

$$T(V_0, E) = \frac{k}{k_0} \frac{|D|^2}{|A_0|^2} = \frac{|D|^2}{|A_0|^2}$$

Limit as $L \gg \frac{1}{k}$ and $\alpha L \gg 1$,
 $e^{-\alpha L} \rightarrow 0$

$$\frac{D}{A_0} = \frac{-4ika}{(\alpha - ik)^2} e^{-(\alpha - ik)L}$$

$$T = \left| \frac{D}{A_0} \right|^2 = \frac{16k^2 \alpha^2 e^{-2\alpha L}}{(\alpha^2 + k^2)^2}$$

$$\text{For } \alpha^2 = \frac{2m}{\hbar^2} (V_0 - E), \quad k^2 = \frac{2mE}{\hbar^2}$$

$$T = 16 \left(\frac{E}{V_0} \right) \left(1 - \frac{E}{V_0} \right) e^{-2\alpha L}$$

Bound. and.

$$\textcircled{1} \psi_I(0) = \psi_{II}(0) \rightarrow A_0 + A = B + C$$

$$\textcircled{2} \psi'_I(0) = \psi'_{II}(0) \rightarrow ikA_0 - ikA = -\alpha B + \alpha C$$

$$\textcircled{3} \psi_{II}(L) = \psi_{III}(L) \rightarrow B e^{-\alpha L} + C e^{\alpha L} = D e^{ikL}$$

$$\textcircled{4} \psi'_{II}(L) = \psi'_{III}(L) \rightarrow -\alpha B e^{-\alpha L} + \alpha C e^{\alpha L} = ik D e^{ikL}$$

$$\text{Gives, } 4ikaA_0 = [(\alpha + ik)^2 e^{-\alpha L} - (\alpha - ik)^2 e^{\alpha L}] D e^{ikL}$$

$$\frac{D}{A_0} = \frac{4ika}{(\alpha + ik)^2 e^{-\alpha L} - (\alpha - ik)^2 e^{\alpha L}} e^{-ikL}$$

8.2, 8.3 3D. Schrodinger Eq.

$$\nabla^2 \psi(x, y, z) = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{2m}{\hbar^2} (V(x, y, z) - E) \psi(x, y, z)$$

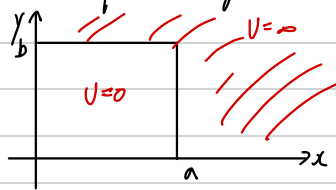
Similarities:

$$\psi(x) \rightarrow \psi(x, y, z) \text{ or } \psi(\vec{r})$$

$$V(x) \rightarrow V(x, y, z) \text{ or } V(\vec{r})$$

$$\frac{\partial^2 \psi(x)}{\partial x^2} \rightarrow \nabla^2 \psi(x, y, z) \text{ or } \nabla^2 \psi(\vec{r})$$

2D Square (rectangular) box



↳ a.) $\psi = 0$ for $x < 0, x > a$ and $y > b, y < 0$

b.) For $x \in [0, a], y \in [0, b]$, use separation of var.

$$V(x, y) = \begin{cases} 0, & x \in [0, a], y \in [0, b] \\ \infty, & \text{elsewhere} \end{cases}$$

① Guess $\psi = X(x)Y(y) \longrightarrow \frac{X''}{X} + \frac{Y''}{Y} = -\frac{2mE}{\hbar^2}$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\frac{2mE}{\hbar^2} \psi(x, y)$$

② $\frac{X''}{X} = -k_x^2, \frac{Y''}{Y} = -k_y^2$

③ $\begin{cases} X(a) = B \sin(k_x a) \\ Y(b) = C \sin(k_y b) \end{cases}, k_x = \frac{n_x \pi}{a}, k_y = \frac{n_y \pi}{b}$, Solutions to 1D infinite well

④ $\psi(x, y) = A \sin\left(\frac{n_x \pi}{a} x\right) \sin\left(\frac{n_y \pi}{b} y\right)$

$$= A \left[\frac{e^{ik_x x} - e^{-ik_x x}}{2i} \right] \left[\frac{e^{ik_y y} - e^{-ik_y y}}{2i} \right]$$

↳ Momentum

$$p_x = \pm \hbar k_x$$

$$p_y = \pm \hbar k_y$$

↳ Energy, $E = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} \right)$

Square

↳ For $a=b$,

↳ Degeneracy

$$E_{21} = E_{12} \text{ etc.}$$

$$E_{n_x, n_y} = \frac{\hbar^2 \pi^2}{2m a^2} (n_x^2 + n_y^2)$$

$$E_{1,1} = \frac{\hbar^2 \pi^2}{m a^2} = 2 E_0$$

8.4 2D Central-Force Problem

↳ $\psi = \psi(r, \phi) \left[\frac{1}{m} \right]$

Schrodinger's eq. gives:

$$\frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \phi^2} = -\frac{2m}{\hbar^2} [U(r) - E] \psi \left[\frac{1}{m^2} \right]$$

① separation of variables, assume $\psi = \Phi(\phi) R(r)$

② multiply $r^2/R\Phi$

③ $\Phi'' = -m^2 \Phi \longrightarrow \Phi(\phi) = e^{im\phi}$

$$R'' + \frac{R'}{r} - \left[\frac{m^2}{r^2} + \frac{2m}{\hbar^2} (U - E) \right] R = 0$$

$$\frac{\Phi''}{\Phi} = \frac{r^2 R'' + r R'}{R} + \frac{2m r^2}{\hbar^2} [U(r) - E] = -m^2$$

↳ Periodicity condition

$$\psi(r, \phi) = \psi(r, \phi + 2\pi)$$

$$\Phi(\phi) = \Phi(\phi + 2\pi)$$

$$\rightarrow m = 0, \pm 1, \pm 2, \dots$$

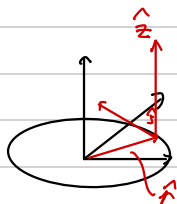
↳ Quantization of Angular Momentum

$$e^{im\phi} = e^{i(m/r)s} \quad \text{--- } s = \text{arc travelled} = r\phi$$

Since $e^{i(m/r)s} \Leftrightarrow e^{ikx}$

Momentum: $\hbar m/r \hat{s} \Leftrightarrow \hbar k \hat{x}$

Angular momentum $L = r \cdot \hat{r} \times \hbar m/r \hat{s}$
 $= \hbar m \hat{z}$
Quantization



↳ Energy levels

- $R(r)$ involves E, m
- Each m has allowed energies $n=1, 2, 3, \dots$
- Thus different energy for different $L = \hbar m, E_{m,n}$
- Degeneracy:
 $E_{m,n} = E_{-m,n}$ since $m^2 = (-m)^2$

8.5 3D Central-Force Problem

↳ $\psi = \psi(\rho, \theta, \phi)$ [$\frac{1}{r^2}$]

Schrodinger's eq. give:

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r\psi) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} = \frac{2M}{\hbar^2} [V(r) - E] \psi$$

① $\Phi''(\phi) = -m^2 \Phi(\phi)$

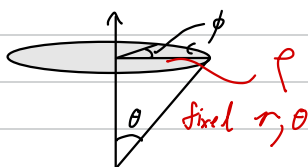
↳ $\Phi(\phi) = e^{im\phi}$

② $\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left(k - \frac{m^2}{\sin^2 \theta} \right) \Theta = 0$

③ $\frac{d^2}{dr^2} (rR) = \frac{2M}{\hbar^2} \left[V(r) + \frac{\hbar^2 k^2}{2Mr^2} - E \right] (rR)$

8.6 Quantization of Angular Momentum

①: $\Phi(\phi) = e^{im\phi}$, $m = 0, \pm 1, \pm 2$
 $= e^{i(m/\rho)s}$, $\phi = s/\rho$



Tangential momentum, $p_{\text{tang}} = \hbar m / \rho$, $m = \mathbb{Z}$

$\hat{=}$ Angular momentum, $L_z = \hbar m$, $m = \mathbb{Z}$

② $\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) + \left(k - \frac{m^2}{\sin^2\theta} \right) \Theta = 0$

Legendre's equation

Solutions when

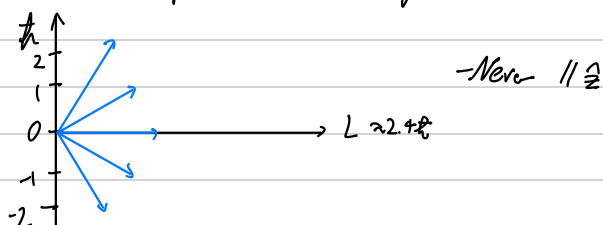
① $k = l(l+1)$

② $l \geq |m| \longrightarrow \Theta_{lm}(\theta)$

Bohr's Quantization

$L = \sqrt{l(l+1)} \hbar \sim \sqrt{l^2} \hbar$ (for large l)

Vector representation of angular momentum.



(example for $l=2$)

8.7 Energy Level of hydrogen

Radial eq.

$\frac{d^2}{dr^2} (rR) = \frac{2M}{\hbar^2} \left[V(r) + \frac{\hbar^2 k^2}{2Mr^2} - E \right] (rR)$

Potential $\leftarrow$ coulomb force const.

$V(r) = -\frac{k_e e^2}{r}$

$\frac{d^2}{dr^2} (rR) = \left(\frac{1}{na_0^2} - \frac{Z}{a_0 r} \right) rR$, for Bohr radius,
 $a_0 = \hbar^2 / m_e k_e e^2$

Solution

$\hookrightarrow$ Exists when

$E = -\frac{n_e (k_e e^2)^2}{2\hbar^2} \cdot \frac{1}{n^2}$

$\hookrightarrow$ No dependence on m

$\hookrightarrow E_H = 13.6 \text{ eV}$

$\hookrightarrow$ Does not matter for direction e^- orbits Hydrogen

$\hookrightarrow$ Thus $E = -\frac{E_H}{n^2}$ for $n=1,2,3$ and $n > l$

$\hookrightarrow$ All $l=2$ $m=\pm 2, \pm 1, 0$ have same energy
 $l=1$ $m=\pm 1, 0$

$\hookrightarrow$ thus $l=0, 1, 2, \dots, (n-1)$

Quantum number and degeneracies

↳ In n^{th} level, n possible values of L exist,

$$\rightarrow l=0, \dots, n-1$$

	s (sharp)	p (principle)	d (diffuse)	f (fundamental)	g h i
quantum number (l)	0	1	2	3	4 5 6

↳ Ground state $\rightarrow 1s$

↳ 1st excited level $\rightarrow 2s, 2p$

↳ Degeneracy

↳ For given n, l

↳ m can have $(2l+1)$ distinct states, $m = l, l-1, \dots, -(l-1), -l$

$$\rightarrow s (l=0) \rightarrow m=0$$

$$\rightarrow p (l=1) \rightarrow m=0, \pm 1$$

$$\rightarrow d (l=2) \rightarrow m=0, \pm 1, \pm 2$$

↳ Total degeneracy for n^{th} level = $\sum$ all numbers in level
 $= n^2$ (before e^- spin)

8.8 Hydrogenic Wave functions

Ground state ($n=1, l=0, m=0$)

$$\rightarrow n=1, l=0, m=0$$

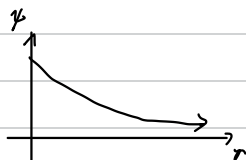
↳ indep. of θ, ϕ , spherically symmetric

↳ Radial eq.

$$\frac{d^2}{dr^2} (rR) = \left(\frac{1}{n^2 a_0^2} - \frac{2}{a_0 r} \right) rR$$

↳ Solution

$$R_{1s}(r) = A e^{-r/a_0}$$



↳ Non zero at $r=0$ for $l=0$

(any $l \neq 0 \rightarrow$ zero at $r=0$)

↳ Classically, a particle can be found at $r=0$ only if $\vec{L}=0$

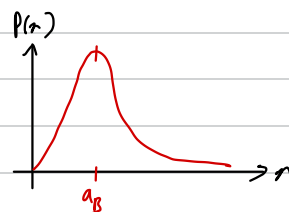
Radial probability density

$$\begin{aligned} \rightarrow dP &= |\psi|^2 dV \\ &= |\psi|^2 4\pi r^2 dr \\ P(r) &= 4\pi r^2 |R(r)|^2 \end{aligned}$$

Probability @ Ground state

$$P_{1s}(r) = 4\pi A^2 r^2 e^{-2r/a_0}$$

↳ Max at $r=a_0$



↳ Potential energy:

$$\langle V \rangle = \int_0^\infty V(r) P(r) dr$$

$$= -\frac{\hbar^2}{a_0}, \text{ agrees with classical prediction}$$

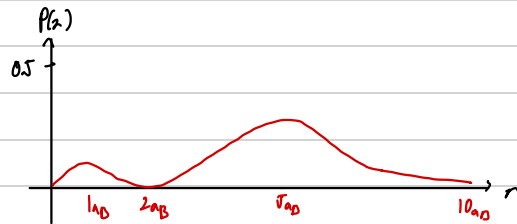
↳ Normalization const.

$$A = \frac{1}{\sqrt{\pi a_0^3}}$$

2s Wave function ($n=2, l=0, m=0$)

↳ Sol. (Symmetric)

$$R_{2s}(r) = A \left(2 - \frac{r}{a_B} \right) e^{-r/2a_B}$$



↳ Probability

max at $5.2 a_B \sim 4 a_B$ ← Second Bohr rad

secondary peak at $0.76 a_B \sim 0 a_B$ ← Not low prob finally it close to nucleus

2p_z Wave function ($n=2, l=1, m=0$)

↳ Wave function

$$\psi(r, \theta, \phi) = R_{2p}(r) \cos \theta$$

↳ Prob.

↳ Depends on θ

↳ max at $z = \pm 2a_B$

↳ Sol.

$$R_{2p}(r) = A r e^{-r/2a_B}$$

↳ 0 at xy plane ($\cos \theta = 0$)

↳ concentrated near z axis (hence 2p_z)

$$\psi = A r e^{-r/2a_B} \cos \theta$$

↳ 0 at $r=0$

2p_x, 2p_y wave functions ($n=2, l=1, m=\pm 1$)

$$\psi_{2p_z} = A r e^{-r/2a_B} \cos \theta = A z e^{-r/2a_B}$$

$$\psi_{2p_x} = A x e^{-r/2a_B}$$

since Schrödinger's eq. involves ∇^2 → concentrated near x/y axis

$$\psi_{2p_y} = A y e^{-r/2a_B}$$

each coordinates in exactly same way

Prob of ^{any} 2p states

For any 2p states

$$P_{2p}(r) = 4\pi r^2 |R_{2p}(r)|^2 = 4\pi A^2 r^4 e^{-r/a_B}$$

↳ max at $r = 4a_B$

Note*: - 2p_x and 2p_y is not exactly $m=1$ or $m=-1$
it is a linear superposition of both

- 2p_z is exactly $m=0$

- All 2p state function is a combination of 3 wave functions with $m=\pm 1, 0$
or of 2p_x, 2p_y, 2p_z

- We chose 2p_x, 2p_y, 2p_z for convenience

8.1 Shells

Most probable radii (shell)	Transitions														
<table border="1"> <thead> <tr> <th>state</th><th>rad</th></tr> </thead> <tbody> <tr> <td>1s</td><td>a_0</td></tr> <tr> <td>2s</td><td>$5.2a_0$</td></tr> <tr> <td>2p</td><td>$4a_0$</td></tr> <tr> <td>3s</td><td>$13.1a_0$</td></tr> <tr> <td>3p</td><td>$12a_0$</td></tr> <tr> <td>3d</td><td>$9a_0$</td></tr> </tbody> </table> $\rightarrow$ reasonably separated from another $\rightarrow$ close to each other	state	rad	1s	a_0	2s	$5.2a_0$	2p	$4a_0$	3s	$13.1a_0$	3p	$12a_0$	3d	$9a_0$	$\sim 0.10\text{eV}$ $\left\{ \begin{array}{l} n=1 \rightarrow n=2 \\ n=1 \rightarrow n=3 \end{array} \right.$ Lyman series 10eV 12eV $\rightarrow$ Only allowed transitions b/w energy levels $\left\{ \begin{array}{l} n=2 \rightarrow n=3 \\ n=2 \rightarrow n=4 \end{array} \right.$ Balmer series 2.0eV 1.5eV $\left\{ \begin{array}{l} n=3 \rightarrow n=4 \\ n=3 \rightarrow n=5 \end{array} \right.$ Paschen series $\sim 0.7\text{eV}$
state	rad														
1s	a_0														
2s	$5.2a_0$														
2p	$4a_0$														
3s	$13.1a_0$														
3p	$12a_0$														
3d	$9a_0$														

8.1b Hydrogen-like atoms

- $\rightarrow$ Single e^- , nucleus charge Ze (not neutral)
- $\rightarrow U = Zke^2/r$

Energy	Radius Parameter	Ground-state
$E = -Z^2 \frac{E_R}{n^2}$	$a_0 \rightarrow a_0/Z$	$\psi_{1s} = A e^{-Zr/a_0}$

- $\rightarrow$ Angular solutions are identical

9.2 Electron spin

Classical spin

- $\rightarrow$ Earth orbits the sun (Orbital angular momentum)
- $\vec{L} = \vec{r} \times \vec{p}$
 - $\vec{r} \rightarrow$ pos. vector rel. to the sun
 - $\vec{p} \rightarrow$ earth's linear momentum

- $\rightarrow$ Earth's own spinning (Spin)
- $\vec{J} = I\vec{\omega}$
 - $I \rightarrow$ moment of inertia
 - $\omega \rightarrow$ angular velocity

- Earth's axis $\nparallel \vec{L}_{\text{orbital}}$
- $\rightarrow$ Total angular momentum, $\vec{J} = \vec{L} + \vec{S}$

Electrons

$$\vec{J}_{\text{elec}} = \vec{L}_{\text{elec}} + \vec{S}_{\text{elec}}$$

$$\rightarrow \text{Recall } |\vec{L}_{\text{elec}}| = \sqrt{l(l+1)}\hbar, l = 0, 1, 2, \dots$$

$$\rightarrow |\vec{S}_{\text{elec}}| = \sqrt{s(s+1)}\hbar, s = 1/2$$

$$= \sqrt{3}/2\hbar, \text{ always same magnitude}$$

$$\rightarrow Z \text{ component of } S,$$

$$S_z = m_s \hbar, m_s = -1/2, 1/2 \text{ (integer + } \hbar \text{ step)}$$

$\downarrow$ spin down $\uparrow$ spin up

Classical analogy

- $\rightarrow$ Electron is pointlike
- $\rightarrow$ No mass/charge distribution (technically cannot spin around axis)
- $\rightarrow$ Spin is an intrinsic property that follows our intuition

Specifying H-Atom state

- $\hookrightarrow n \rightarrow$ energy level, $E = -\frac{E_0}{n^2}$
 $\hookrightarrow l \rightarrow$ Angular momentum, $L = \sqrt{l(l+1)}\hbar$, $l < n$
 $\hookrightarrow m_l \rightarrow$ z Angular momentum, $L_z = m_l \hbar$, $l \geq |m_l|$
 $\hookrightarrow m_s \rightarrow$ z spin direction, $S_z = m_s \hbar$, $m_s = \pm \frac{1}{2}$

Spatial (Orbital)
distribution

Degeneracy $\rightarrow 2n^2$

Beyond e^-

- $\hookrightarrow$ All other sub-atomic particles have spin
 $\hookrightarrow$ Not all have $\bar{s} = \frac{1}{2}$

Particle	spin
e^-	$\frac{1}{2}$
p^+	$\frac{1}{2}$
n	$\frac{1}{2}$
Neutrino, ν	$\frac{1}{2}$
Omega, Ω^-	$\frac{3}{2}$
Pion, π^0	0
Alpha particle, α	0
Photon, γ	1
Deuteron, d	1
Graviton	2

Half
Integral
Spin

Integral
Spin

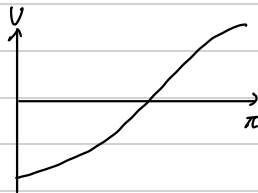
9.3 Magnetic moments

- $\hookrightarrow$ Current i flowing in plane loop A in magnetic field $\vec{B}$ experiences

Torque, $\vec{\tau} = i \vec{A} \times \vec{B}$, $\vec{A}$ = Area vector, normal to area plane
 $= \vec{\mu} \times \vec{B}$

- $\hookrightarrow \vec{\mu}$ is called magnetic moment of the loop

$\hookrightarrow$ Work, $W = - \int \tau d\theta$
 $= -\mu B \int \sin\theta d\theta$
 $= \mu B \cos\theta$



- $\hookrightarrow$ Potential, $U = -W = -\vec{\mu} \cdot \vec{B}$

Current loop generated by e^- in orbit r , period T

↳ Current = Total charge passing any fixed point in unit time

$$i = \frac{e}{T} = e \frac{v}{2\pi r}$$

↳ Magnetic moment,

$$\mu = iA = \frac{ev}{2\pi r} \pi r^2 = \frac{1}{2} evr$$

↳ Angular momentum,

$$\vec{L} = m_e v r$$

↳ Gyromagnetic ratio

$$\frac{\mu}{L} = \frac{e}{2m_e}, \text{ depends only on charge, mass of } e^-$$

$$\text{Thus } \vec{\mu} = -\frac{e}{2m_e} \vec{L}$$

q.t (Normal) Zeeman effect

Helium atom

↳ Sph of 2 e^- in opposite directions

↳ Cancels out (Singlet states)

↳ w/o a magnetic field,

$\vec{L}$ has $2l+1$ possible orientations,

$2l+1$ values of $L_z = m_l \hbar$

E_0 $2l+1$ fold-degenerate

Magnitude of shift

↳ Shifts are $\sim 10^{-4}$ of absorbing energy levels

↳ Freq. of light for transition from first excited state to ground

$$f_0 = E_r/h = E_0/h$$

↳ Triplet in magnetic field

$$f_0, f_0 + \mu_B/h, f_0 - \mu_B/h$$

(Only, and always triplets)

↳ w/ magnetic field

$$E = E_0 + \Delta E$$

$$= E_0 - \vec{\mu} \cdot \vec{B}$$

$$= E_0 + \left(\frac{e \hbar}{2m_e} \right) m_l B$$

$$= E_0 + m_l \mu_B B$$

$m_l \rightarrow$ magnetic quantum number

$$\mu_B = \frac{e \hbar}{2m_e} = 9.27 \times 10^{-24} \text{ A} \cdot \text{m}^2, \text{ Bohr magneton}$$

$$= 5.79 \times 10^{-5} \frac{\text{eV}}{\text{T}} \quad \text{Magnetic moment units}$$

9.5 Spin Moment

Spin magnetic Moment

$\vec{\mu}_{\text{spin}} = -\frac{e}{m_e} \vec{S}$ "e⁻ spinning on its axis"
experimentally determined

↳ Classical e⁻ radius, $\frac{1}{4\pi\epsilon_0} \frac{q^2}{r_e} = m_e c^2$, $r_e = 3 \times 10^{-16} \text{ m}$
e⁻ are still "dots", not spinning spheres

Total magnetic moment

$\vec{\mu}_{\text{atom}} = \vec{\mu}_{\text{orbital}} + \vec{\mu}_{\text{spin}}$
 $= -\frac{e}{2m_e} (\vec{L} + 2\vec{S})$

9.6 Anomalous Zeeman Effect

Hydrogen Fine Structure

↳ Consider ground state of H, $1s$, $\vec{L}=0$

$\vec{\mu}_{\text{orb}} = \vec{\mu}_{\text{spin}} = -\frac{e}{m_e} \vec{S}$

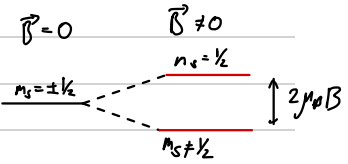
↳ Then in $\vec{B}$,

$\Delta E = -\vec{\mu} \cdot \vec{B} = \frac{e}{m_e} S_z B$ and $S_z = \pm \frac{1}{2} \hbar$

↳ Thus $\Delta E_z = \pm \frac{e \hbar}{2m_e} B = \pm \mu_B B$
bohr magneton

↳ Two levels separated by $2\mu_B B$
 $\approx 1.2 \times 10^{-4} \text{ eV} \cdot \frac{B}{\text{T}}$ "small shift"

↳ Only 2 levels instead of $2l+1$



10.2 Independent Particle Approximation

IPA

↳ Each electron interact by force from nucleus of charge Ze

and other e⁻ of charge $-(Z-1)e$

↳ Treat motion of each e⁻ independently, consider force from nucleus +

force of static distribution for $Z-1$ other e⁻

↳ This potential is $V_{\text{IPA}}(r)$

↳ Central Field Approx: The further assumption that the $Z-1$ other e⁻ are spherically symmetric around nucleus

$V_{\text{IPA}}(r)$

↳ Charge Q inside radius r , $V(r) = -kQe/r$

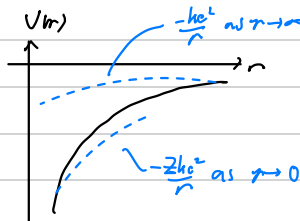
↳ 1st e⁻, closest to nucleus, $V(r) = -\frac{kZe^2}{r}$

↳ 2nd e⁻, $V(r) = -\frac{k(Z-1)e^2}{r}$

;

↳ Last e⁻, $V(r) = -\frac{ke^2}{r}$

↳ Thus, $V(r) \approx -Z_{\text{eff}}(r) \frac{ke^2}{r}$



10.3 IPA Energy Levels

Energy Levels

↳ Using $V(r) = -Z_{\text{eff}}(r) \frac{ke^2}{r}$, Ψ can still be separated into $R(r)$, $\Theta(\theta)$, $\Phi(\phi)$
identical to before

↳ Energy levels $E_n (\neq \frac{E_n}{n^2})$

1st and 2nd e⁻

↳ Go to $1s$ state

$E_1 \approx -Z^2 E_R$

3rd and subsequent e⁻

↳ Go to higher shells (Pauli exclusion)

↳ 2s has same P(r) almost as near as 1s

↳ P_{2s}^{2PA}(r) extends to lower r → More binding energy

↳ e⁻ see Z^{eff} → Z^{TOT}

↳ P_{2p}^{2PA}(r) rise at higher r → Less binding energy

↳ e⁻ see Z^{eff} → Z^{TOT} - 2

★ U_{ZPA} biases binding energy for higher angular momentum states

$$Z_{\text{eff}}^s > Z_{\text{eff}}^p > Z_{\text{eff}}^d > \dots$$

$$E^{ns} < E^{np} < E^{nd} < \dots$$

4s

↳ At some point, the ns level is below the (n-1)d level

↳ Thus 4s level is less tightly bound than 3d

Most Probable Radius, r_{mp}

↳ For H-like atoms: ↳ For other atoms:

$$r_{mp} = n^2 a_0 / Z$$

$$r_{mp} = n^2 a_0 / Z_{\text{eff}}(r)$$

↳ Z^{eff} decreases as n gets larger → r_{mp} increases for higher # of e⁻

10.4 Pauli's Exclusion Principle

↳ No two e⁻ in a quantum system can occupy the same quantum state

Building Neutral Atoms

Z=1 (Hydrogen)

↳ e⁻ in 1s, spin up or down

$$\hookrightarrow E_{\text{ion}} = 13.6 \text{ eV}$$

$$\hookrightarrow \vec{\mu} = \vec{\mu}_{\text{spin}}$$

Z=2 (Helium)

↳ 2 e⁻ in 1s, ↑↓ one up, one down

↳ Z^{eff} ≈ 1.4 "1st e⁻ partially screens 2nd one"

$$\hookrightarrow \vec{\mu} = \vec{\mu}_{\text{spin}} = 0$$

$$\hookrightarrow E_{\text{ion}} = 24.6 \text{ eV} < (Z=2)^2 E_H$$

$$\approx (Z_{\text{eff}})^2 E_H$$

~ 2x harder to ionize than H atom

↳ 1st electron has E_{ion} ≈ 50 eV

↳ Very chemically stable → Superfluidity

↳ Transits states at low T ~ 1.7 K

↳ Filled shell (both 1s states occupied)

Z=3 (Lithium)

↳ 2e⁻ in 1s, n=1, l=0, m=0, m_s=±1/2, L+S=0

↳ 3rd e⁻ in 2s, Z^{eff} ≈ 1

$$\hookrightarrow U_{\text{eff}} = Z_{\text{eff}}^2 \frac{m_e c^2}{n}$$

$$\hookrightarrow E_{\text{ion}} = 5.4 \text{ eV} \sim E_{\text{hydrogen}, n=2} = 3.4 \text{ eV}$$

since P_{2s}^{Li}(r) < P_{2s}^H(r) due to some

2s e⁻ prob. closer to Lithium nucleus

shielding at 1st, 2nd e⁻ not quite complete

1.2 Ideal Gas Law

Ideal Gas Law

$$PV = nRT$$

or

$$PV = NkT$$

↳ Ideal gas const.

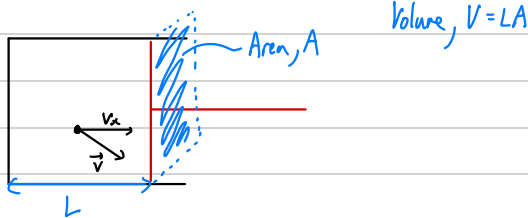
$$R = 8.31 \text{ J/mol K}$$

↳ Number of molecules, $N = N_A \cdot n = (6.02 \times 10^{23})n$

↳ Boltzmann const., $k = \frac{R}{N_A} = 1.381 \times 10^{-23} \text{ J/K}$

$$nR = Nk$$

Microscopic Model



$$\begin{aligned} \textcircled{1} \text{ Pressure}_{\text{avg}}, \bar{P} &= \frac{\bar{F}_x \text{ on piston}}{A} = \frac{-\bar{F}_x \text{ on molecule}}{A} \\ &= -m \left(\frac{\Delta v_x}{\Delta t} \right) \cdot \frac{1}{A} \end{aligned}$$

② Let Δt = time for molecule to travel from left to right and back again (round trip)

$$\Delta t = \frac{2L}{v_x}$$

Molecules undergo 1 collision w/ piston, velocity change:

$$\Delta v_x = v_{x,\text{final}} - v_{x,\text{initial}} = (-v_x) - (v_x) = -2v_x$$

$$\textcircled{3} \bar{P} = -m \left(\frac{-2v_x \cdot v_x}{2L} \right) \frac{1}{A} = \frac{mv_x^2}{LA} = \frac{mv_x^2}{V} \text{ or}$$

$$\bar{P}V = mv_x^2$$

④ For N molecules,

$$\begin{aligned} \bar{P}V &= mv_{x1}^2 + mv_{x2}^2 + \dots \\ &= Nmv_x^2 \end{aligned}$$

⑤ Ideal Gas Law:

$$PV = NkT \longrightarrow Nmv_x^2 = NkT$$

$$\frac{1}{2}mv_x^2 = \frac{1}{2}kT$$

⑥ Root-mean-square velocity

$$\hookrightarrow \sqrt{\bar{v}^2} \neq v_{\text{avg}}$$

$$\hookrightarrow \bar{v}^2 = \frac{3kT}{m}$$

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

For 3D systems, translational kinetic energy,

$$K_{\text{trans}} = \frac{1}{2}Nmv_x^2 + \frac{1}{2}Nmv_y^2 + \frac{1}{2}Nmv_z^2 = \frac{3}{2}NkT$$

1.3 Equipartition Theorem

Theorem

At temp T , avg energy of any quadratic deg. of freedom (deg.) is $\frac{1}{2}kT$

If system has N particles,
 f deg.,

Then Thermal Energy, $U_{\text{thm}} = N \cdot f \cdot \frac{1}{2}kT$

↳ Technically it is U_{thm} , but for $N \gg$, $U_{\text{avg}} \approx U_{\text{thm}}$

Degree of Freedom

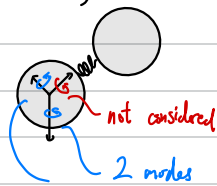
Consider diatomic molecules

↳ Translational, 3 axes

$$E_{\text{trans}} = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$$

- deg. = 3

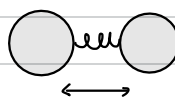
↳ Rotational, for diatomic molecules



$$E_{\text{rot}} = \frac{1}{2}I\omega_y^2 + \frac{1}{2}I\omega_z^2$$

- deg. = 2

↳ Vibrational (Only at high temp)



1D vibration

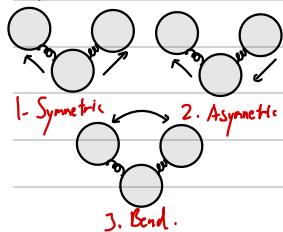
$$E_{\text{vib}} = \frac{1}{2}k(r_1 - r_2)^2$$

- deg. = 1

- For vibration, each deg. contributes $k_B T$ instead of $\frac{1}{2}k_B T$

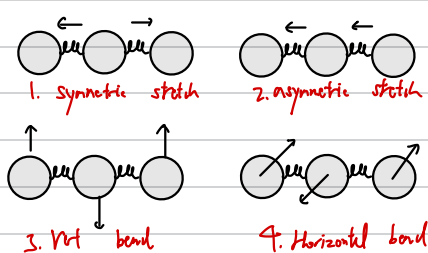
- Since each mode $\rightarrow$ movement of 2 atoms $\rightarrow$ potential energy in bond

↳ Water vibration modes



In general,

↳ CO₂ vibration modes



↳ Solids



- deg. ≈ 6

3 from kinetic energy +
3 from potential in spring

↳ Liquid

$$KE = \frac{1}{2}mv^2$$

- but equipartition theorem does not hold,
non quadratic potential

Linearity	Trans.	Rot.	Vib.	Total
Yes	3	2	$3n - 5$	$3n$
No	3	3	$3n - 6$	$3n$

1.4 Heat, Work

Q , Heat $\rightarrow$ Flow of energy from 1. object to another due to temperature difference

W , Work $\rightarrow$ Any other energy transfer

$$\Delta U = Q + W$$

By conduction, convection, radiation

1.5 Compression work

Quasistatic process

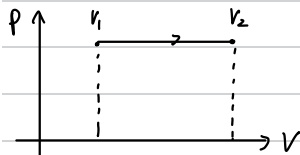
→ A slow process where the system is constantly in equilibrium

$$W = - \int_{V_1}^{V_2} P dV$$

→ Isobaric

- Const. P

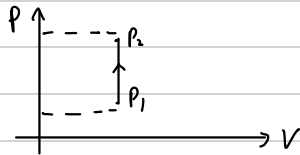
$$W = -P(V_2 - V_1)$$



→ Isochoric

- Const V

$$W = 0$$



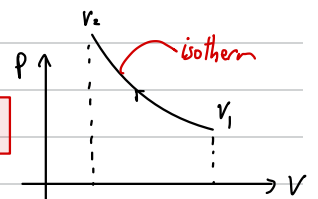
→ Isothermal

→ Const $T \rightarrow P = NkT/V$

$$W = -NkT \int \frac{dV}{V} = NkT \ln\left(\frac{V_2}{V_1}\right)$$

→ Since $U = NkT/2$, $\Delta U = 0$

$$\rightarrow Q = NkT \ln\left(\frac{V_2}{V_1}\right)$$



Adiabatic process

→ Process so fast s.t. no heat exchange $\rightarrow Q = 0$

→ Can be quasistatic or not

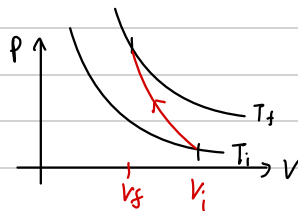
$$\Delta U = Q + W$$

$$f \cdot N \cdot k \frac{dT}{2} = -P dV$$

$$f \cdot N \cdot k \frac{dT}{2} = - \frac{NkT}{V} dV$$

$$\frac{f}{2} \frac{dT}{T} = - \frac{dV}{V}$$

$$\begin{cases} VT^{f/2} = \text{const} \\ V^{\gamma} P = \text{const}, \gamma = \frac{f+2}{f} \end{cases}$$



1.6 Heat capacity, Latent Heat, Enthalpy

Heat capacity

→ Heat req. to raise an object's temperature per degree of increase

→ Const. vol.

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

→ Const. pressure

$$C_P = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P$$

→ Quasistatic systems

→ Ideal gas

→ Generally,

$$C_V = \frac{Nk f}{2} = \frac{Nk f}{2}$$

$$\left(\frac{\partial U}{\partial T} \right)_P = \frac{Nk f}{2} + Nk \left(\frac{\partial V}{\partial T} \right)_P, C_P = C_V + Nk \left(\frac{\partial V}{\partial T} \right)_P$$

$$C_P = Nk \left(\frac{f}{2} + 1 \right)$$

Latent heat

→ For phase change, $C = \frac{Q}{\Delta T} = \frac{Q}{0} \rightarrow \infty$

→ Thus L = latent heat, energy req. for phase change,
 $l \equiv L/m$, specific latent heat

Enthalpy

$$H \equiv U + PV$$

→ Hess' Law

$$\Delta H = \Delta U + P \Delta V \text{ for const } P$$

$$\Delta H = \sum n \Delta H_f^\circ, \text{product} - \sum m \Delta H_f^\circ, \text{reactants}$$

= $Q + W_{\text{other}}$, ignore compression-expansion work

$$\rightarrow C_P = \left(\frac{\partial H}{\partial T} \right)_P$$

2.1 Two-state systems

For a system of N items, multiplicity of macrostate with n "heads" is:

$$\Omega(N, n) = \binom{N}{n} = \frac{N!}{(N-n)! n!}$$

2-state paramagnet

↳ Particles act like compass needles that tend to align parallel to external magnetic field

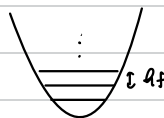
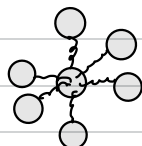
↳ $QM \rightarrow 2$ values $\rightarrow 2$ -states $N_\uparrow, N_\downarrow$



$$\Omega(N, N_\uparrow) = \frac{N!}{(N-N_\uparrow)! N_\uparrow!} = \frac{N!}{N_\downarrow! N_\uparrow!}$$

2.2 Einstein Model of a Solid

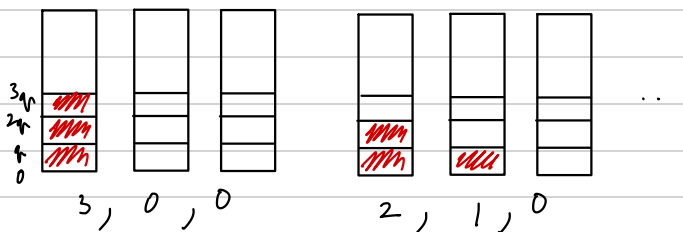
Einstein model



↳ Each oscillator has evenly spaced energy levels, and same frequency

$$E = (q + \frac{1}{2}) \hbar \sqrt{\frac{k}{m}}$$

↳ Ex. For 3 oscillators w/ $q_{\text{total}} = 3$



In total, $\Omega(3) = 10$

↳ Generally, for q quanta,

→ Multiplicity:

$$\Omega(N, q) = \binom{q+N-1}{q} = \frac{(q+N-1)!}{q! (N-1)!}$$

→ Probability:

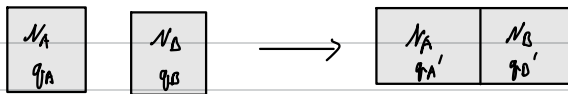
$$p = \frac{1}{\Omega(N, q)}$$

2.3 Interacting Systems

↳ Assume weakly coupled

(exchange of energy b/w solid a lot slower than exchange of energy with each solid)

In short the scale, individual energy is $\sim$ constant



$$N_A + N_B = N$$

$$q_A + q_B = q_A' + q_B' = q$$

Multiplicity of 2 solids

$$\Omega_{\text{total}} = \Omega_A \cdot \Omega_B$$

Fundamental assumption of statistical mechanics

In an isolated system in thermal equilibrium, all accessible microstates are equally probable

2.4 Large systems

Large Systems

$$\textcircled{1} \text{ Multiplicity } \Omega(N, q) = \frac{(q+N-1)!}{q! (N-1)!} \approx \frac{(q+N)!}{N! q!}$$

$$\textcircled{2} \text{ Stirling's approximation } \rightarrow N! \approx N^N e^{-N} \sqrt{2\pi N} \approx N^N e^{-N} \text{ for large } N$$

$$\ln N! \approx \ln(N^N e^{-N}) = N \ln N - N$$

$$\textcircled{3} \ln[\Omega(N, q)] \approx (q+N) \ln(q+N) - N \ln N - q \ln q$$

$$\textcircled{4} \text{ For } q \gg N, \ln(q+N) \approx \ln q + \frac{N}{q}$$

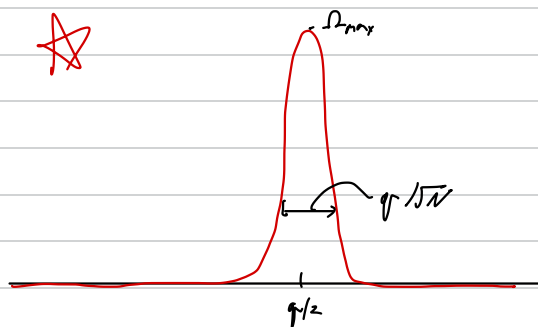
$$\ln \Omega \approx (\ln q + \frac{N}{q})(q+N) - N \ln N - q \ln q$$

$$= N \ln q + \frac{N}{q}(q+N) - N \ln N$$

$$= N \ln\left(\frac{q}{N}\right) + N + \cancel{\frac{N^2}{q}} \rightarrow 0$$

$$\textcircled{5} \ln \Omega = N \left[\ln\left(\frac{q}{N}\right) + \ln e \right]$$

$$\Omega = \left(\frac{eq}{N}\right)^N \text{ for } q \gg N$$



2 Large Systems

$$\hookrightarrow \Omega = \left(\frac{eq_A}{N}\right)^N \left(\frac{eq_B}{N}\right)^N = \left(\frac{e}{N}\right)^{2N} (q_A q_B)^N$$

↳ Peak at

$$\Omega_{\text{max}} = \left(\frac{e}{N}\right)^{2N} \left(\frac{q}{2}\right)^{2N}$$

Sharpness of multiplicity

$$\textcircled{1} \text{ Let } q_1 = \frac{q}{2} + x, \quad q_2 = \frac{q}{2} - x \text{ (near peak)}$$

$$\textcircled{2} \text{ Then } \Omega = \left(\frac{e}{N}\right)^{2N} \left(\left(\frac{q}{2}\right)^2 - x^2\right)^N$$

$$\textcircled{3} \ln \left[\left(\left(\frac{q}{2}\right)^2 - x^2\right)^N \right] \rightarrow \approx N \left[\ln\left(\frac{q}{2}\right)^2 - \left(\frac{2x}{q}\right)^2 \right]$$

$$\textcircled{4} \Omega \approx \left(\frac{e}{N}\right)^{2N} \left(\frac{q}{2}\right)^{2N} \cdot e^{-N(2x/q)^2}$$

★ Falls by $1/e$ at $x = \frac{q}{2\sqrt{N}}$

Paramagnets

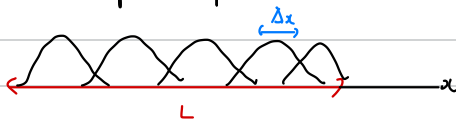
$$\Omega = \binom{N}{n} = \frac{N!}{n!(N-n)!} \approx \left(\frac{N}{n}\right)^n$$

2.5 Ideal Gas

1D Case

↳ Heisenberg's uncertainty Principle gives:
 $\Delta x \Delta p_x \geq \hbar$

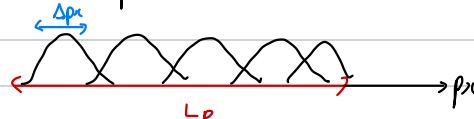
↳ Total possible positions:



Position Space

$$N_x = \frac{L}{\Delta x}$$

↳ Total possible momentums:



Momentum Space

$$N_x = \frac{L_p}{\Delta p_x}$$

$$\Omega = \frac{L}{\Delta x} \frac{L_p}{\Delta p_x} \approx \frac{L p_x}{\hbar}$$

↳ Generalize to $\Omega_{3D} \sim \frac{V_x V_{momentum}}{\hbar^3}$

Monatomic Ideal Gas

$$\begin{aligned} \hookrightarrow U &= \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) \rightarrow p_x^2 + p_y^2 + p_z^2 = \sqrt{2mU} \\ \hookrightarrow V_{momentum} &= \frac{4}{3} \pi (2mU)^{3/2} \quad \text{radius of sphere} \end{aligned}$$

$$\hookrightarrow V_{space} = XYZ$$

$$\hookrightarrow \Omega = \frac{4}{3} \pi (2mU)^{3/2} XYZ \cdot \frac{1}{\hbar^3}$$

2 Distinguishable molecules

2 indistinguishable molecules

$$\hookrightarrow U = \frac{1}{2m} (p_{x1}^2 + p_{y1}^2 + p_{z1}^2 + p_{x2}^2 + p_{y2}^2 + p_{z2}^2)$$

↳ Same U, n

$$\hookrightarrow V_{momentum} = \text{surface area of 6D hypersphere, } r = \sqrt{2mU}$$

$$\hookrightarrow \text{But } \Omega = \frac{V_{space} V_{momentum}^2}{\hbar^6}$$

$$\hookrightarrow \Omega = \frac{V_{space} V_{momentum}^2}{\hbar^6}$$

2 $\hbar^6$
 indistinguishable correction factor

N - indistinguishable molecules

Hypersphere area

Ideal gas

2 same Ideal Gases

$$\hookrightarrow \Omega_N = \frac{1}{N!} \frac{V^N}{\hbar^{3N}} \times \text{Hypersphere Area}$$

$$\hookrightarrow A = \frac{2\pi^{d/2}}{(\frac{d}{2}-1)!} r^{d-1}$$

$$\hookrightarrow \Omega(U, V, N) = f(N) V^N U^{3N/2}$$

$$\hookrightarrow \Omega = f^2(N) (V_A V_B)^N (U_A U_B)^{3N/2}$$

$$\begin{aligned} \hookrightarrow \text{peak width} &= \frac{U_{tot}}{\sqrt{3N/2}} \\ &= \frac{U_{tot}}{\sqrt{N}} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{N!} \frac{V^N}{\hbar^{3N}} \cdot \frac{2\pi^{3N/2}}{(\frac{3N}{2}-1)!} (\sqrt{2mU})^{3N-1} \\ &\approx \frac{1}{N!} \frac{V^N}{\hbar^{3N}} \frac{\pi^{3N/2}}{(3N/2)!} (\sqrt{2mU})^{3N} \end{aligned}$$

2.6 Entropy

$$\hookrightarrow S = k \ln \Omega$$

Properties:

$$\hookrightarrow S_{\text{tot}} = S_1 + S_2$$

$\hookrightarrow$ Entropy is higher when energy is equally distributed

$\hookrightarrow$ Scales large systems with $\ln(x)$

Second Law of thermodynamics

Heat always flows from hotter to colder regions of a system

or

A system in equilibrium will be found in the microstate with the greatest entropy

or

Entropy of an isolated system will increase

Entropy of ideal gas

$$S = Nk \left[\ln \left(\frac{V}{N} \left(\frac{4\pi m U}{3Nk^2} \right)^{3/2} \right) + \frac{5}{2} \right] \quad \star \text{ For indistinguishable gases}$$

Sackur - Tetrode Equation

S increases when:

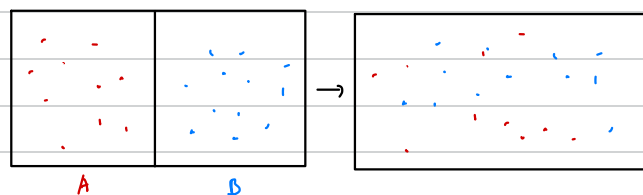
- $\hookrightarrow$ ① Higher Volume $\rightarrow$ More spatial organization possibilities
- $\hookrightarrow$ ② More particles $\rightarrow$ More combinatorial organization possibilities
- $\hookrightarrow$ ③ More energy $\rightarrow$ More energy redistribution possibilities

$$\hookrightarrow \Delta S = S_f - S_i$$

Free expansion, $Q=0$

$$\Delta S = Nk \ln \frac{V_f}{V_i}$$

Mixing (Ex. 2 equal volumes)



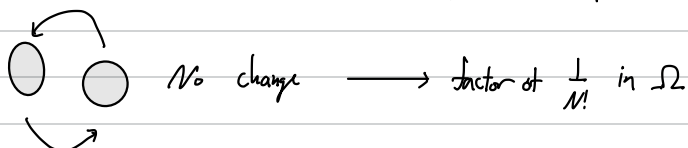
$$\Delta S_A = Nk \ln 2$$

$$\Delta S_B = Nk \ln 2$$

$$\Delta S_A + \Delta S_B = 2Nk \ln 2$$

Gibbs paradox

Identical Gas molecules cannot trade places to produce multiplicity



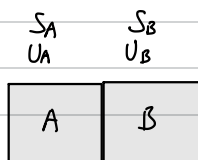
Side: Distinguishable

$$S = Nk \left[\ln \left[V \left(\frac{4\pi m U}{3Nk^2} \right)^{3/2} \right] + \frac{3}{2} \right]$$

3.1 Temperature

Definition

For 2 solids,



Properties

↳ As $T \rightarrow \infty$, $\frac{dS}{dU} \rightarrow 0$

↳ Small gain in temperature results in huge gain in entropy
the system is "cold"

At equilibrium, $S = S_{\max}$

$$\frac{dS_{\text{total}}}{dU_A} = 0$$

$$\frac{dS_A}{dU_A} + \frac{dS_B}{dU_B} = 0, \text{ but } dU_A = -dU_B$$

Thus, $\frac{dS_A}{dU_A} = \frac{dS_B}{dU_B}$ at equilibrium

$$\text{Temperature, } T \equiv \left(\frac{dS}{dU} \right)^{-1} \Big|_{N,V}$$

3.2 Entropy and heat

Heat capacity, again

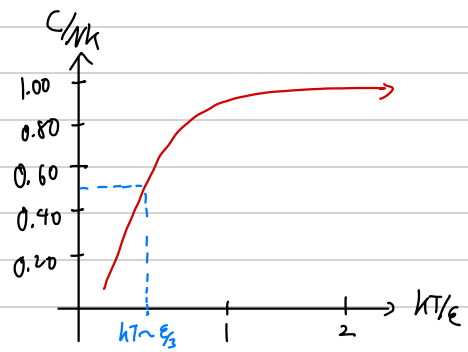
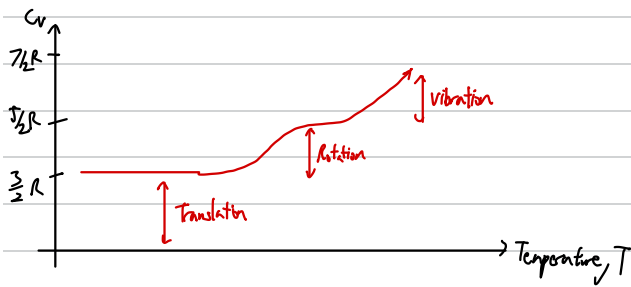
$$c_v = \left(\frac{dU}{dT} \right)_v$$

$$= \left(\frac{dU}{dS} \cdot \frac{dS}{dT} \right)_v$$

$$= T \frac{dS}{dT} \Big|_v$$

$$c_v \rightarrow 0 \text{ for } T \rightarrow 0$$

3.4 Mechanical Equilibrium and Pressure



$\hookrightarrow \epsilon$, Energy of a single quantum of vibration, a d.o.f contributes to C_v
 $\hookrightarrow$ Energy required to activate the vibrational mode

$$\epsilon \approx 3kT$$

Heat capacity $\sim \frac{1}{2}Mk$ at $\epsilon = 3kT$

Pressure - Entropy relation

Consider $0 = \frac{\partial S_{\text{total}}}{\partial V_A}$

U_A, S_A, V_A	U_B, S_B, V_B
-----------------	-----------------

$$\frac{\partial S_A}{\partial V_A} + \frac{\partial S_B}{\partial V_A} = 0$$

$$\frac{\partial S_A}{\partial V_A} = -\frac{\partial S_B}{\partial V_B}, \text{ for } dV_A = -dV_B, \text{ at equilibrium}$$

$$P = T \left(\frac{\partial S}{\partial V} \right)_{U, N}$$

Thermodynamic identity

$$(1) dS = \left(\frac{\partial S}{\partial U} \right) dU + \left(\frac{\partial S}{\partial V} \right) dV + \left(\frac{\partial S}{\partial N} \right) dN \quad \text{for conv. molecules}$$

$$(2) dU = dQ + dW$$

$$(3) \text{ For reversible processes, } dQ = T dS$$

$$(4) dW = -P dV$$

$$\text{Then, } dU = T dS - P dV \quad \text{or} \quad dS = \frac{dU}{T} + \frac{P}{T} dV$$

Entropy, Heat

$$Q = T dS \text{ for quasi-static}$$

$$Q = 0 \text{ for adiabatic, quasi-static}$$

isentropic

$$dS > \frac{Q}{T} \quad \text{when } W > -P dV \quad (\text{entropy generated})$$

3.5 Diffuse Equilibrium and Chemical Potential

Chemical Potential

U_A, N_A, S_A	U_B, N_B, S_B
-----------------	-----------------

$\mu \rightarrow$ How much a system's energy change when you add 1 particle, holding entropy, volume fixed

At equilibrium,

$$\frac{\partial S_A}{\partial N_A} = \frac{\partial S_B}{\partial N_B}$$

Define chemical potential, $\mu \equiv -T \left(\frac{\partial S}{\partial N} \right)_{U,V} = \left(\frac{\partial U}{\partial N} \right)_{S,V}$

Then, $dU = TdS - PdV + \mu dN$
 $dS = \frac{1}{T}dU + \frac{P}{T}dV - \frac{\mu}{T}dN$

Ideal Gas chemical potential

① $S = Nk \left[\ln \left(\frac{V}{N \lambda^3} \right) + \frac{5}{2} \right]$, $\lambda = \frac{h}{\sqrt{2\pi m k T}}$, Thermal de Broglie wavelength

② $\frac{\partial S}{\partial N} = k \ln \left(\frac{V}{N \lambda^3} \right) + \frac{5}{2}k$

$\rightarrow$ Size of a particle's wavepacket
 $\lambda^3 \rightarrow$ volume a particle effectively occupies

③ $\mu = -T \frac{\partial S}{\partial N} = -kT \left[\ln \left(\frac{V}{N \lambda^3} \right) + \frac{5}{2} \right]$

$\rightarrow \frac{V}{N \lambda^3} \rightarrow$ Volume per particle
 Quantum volume

$\rightarrow N/V$ (density) increase $\rightarrow \mu$ increase

$\rightarrow$ For ideal gas, μ is typically negative, T increase $\rightarrow \mu$ more negative

Equilibrium (Summary)

$$\Delta S_{tot} = \left(\frac{1}{T_A} - \frac{1}{T_B} \right) \Delta U_A + \left(\frac{P_A}{T_A} - \frac{P_B}{T_B} \right) \Delta V_A - \left(\frac{\mu_A}{T_A} - \frac{\mu_B}{T_B} \right) \Delta N_A$$

$\frac{\partial S}{\partial U}|_{U,V,N}$ $\frac{\partial S}{\partial V}|_{U,V,N}$ $-\frac{\partial S}{\partial N}|_{U,V}$

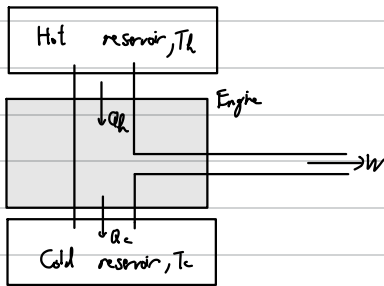
$\rightarrow T_A = T_B \rightarrow$ Thermal equilibrium

$\rightarrow P_A = P_B \rightarrow$ Mechanical equilibrium

$\rightarrow \mu_A = \mu_B \rightarrow$ Chemical equilibrium

4.1 Heat Engine

Heat engine



$$\hookrightarrow \text{Efficiency, } e = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h}$$

$$\text{Since } \Delta S_{\text{tot}} = \Delta S_h + \Delta S_c \geq 0$$

$$\frac{-Q_h}{T_h} + \frac{Q_c}{T_c} \geq 0$$

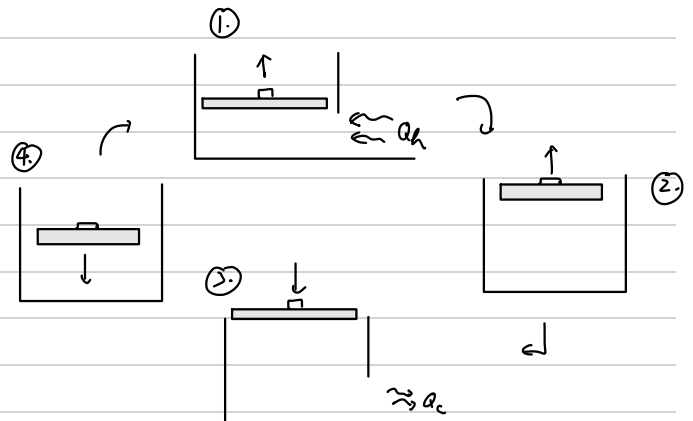
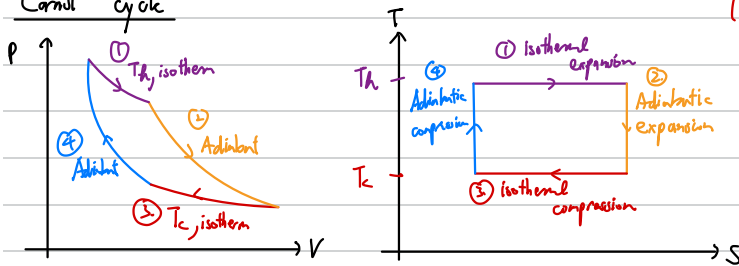
expels absorbs

$$\frac{Q_c}{Q_h} \geq \frac{T_c}{T_h}$$

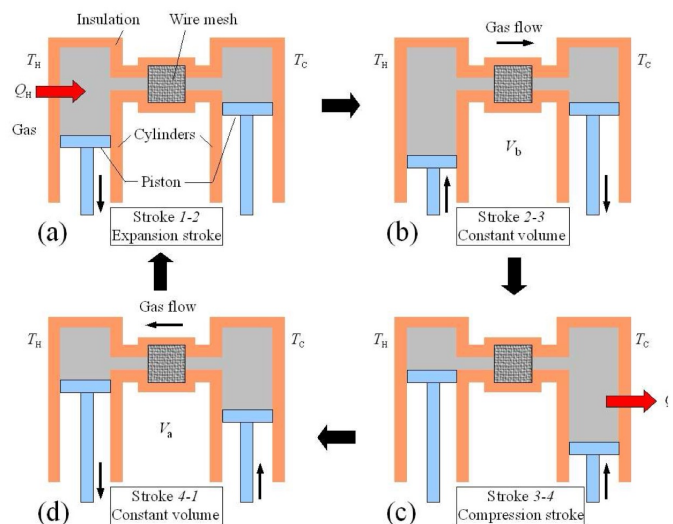
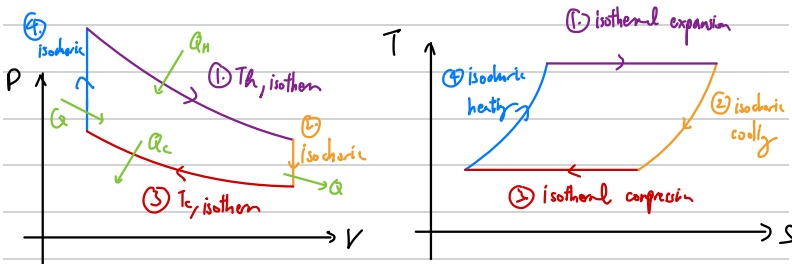
$$\text{Then } e \leq 1 - \frac{T_c}{T_h}$$

Carnot efficiency

Carnot cycle



Stirling Cycle (Engine)



$$Q_1 = NkT_h \ln\left(\frac{V_2}{V_1}\right)$$

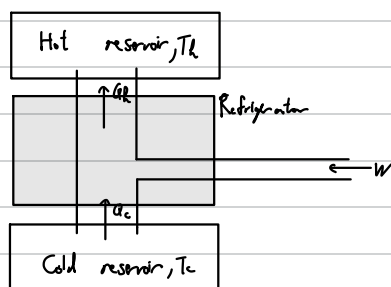
$$Q_c = NkT_c \ln\left(\frac{V_2}{V_1}\right) \rightarrow W = \sum Q_h = Nk(T_h - T_c) \ln\left(\frac{V_2}{V_1}\right)$$

$$Q_3 = -NkT_c \ln\left(\frac{V_2}{V_1}\right)$$

$$Q_4 = Nk(T_h - T_c)$$

$$\hookrightarrow \text{Stirling efficiency, } \eta = \frac{W}{Q_{\text{in}}} = \frac{Nk(T_h - T_c) \ln\left(\frac{V_2}{V_1}\right)}{NkT_h \ln\left(\frac{V_2}{V_1}\right) + Nk(T_h - T_c)} = \frac{(1 - \frac{T_c}{T_h}) \ln\left(\frac{V_2}{V_1}\right)}{\ln\left(\frac{V_2}{V_1}\right) + \ln\left(1 - \frac{T_c}{T_h}\right)} < 1 - \frac{T_c}{T_h}$$

4.2 Refrigerators



$$\hookrightarrow \text{Coefficient of performance, } \text{COP} = \frac{Q_c}{W} = \left(\frac{Q_h}{Q_c} - 1\right)^{-1}$$

$$\text{Since } \Delta S_h + \Delta S_c \geq 0$$

$$\frac{Q_h}{T_h} - \frac{Q_c}{T_c} \geq 0$$

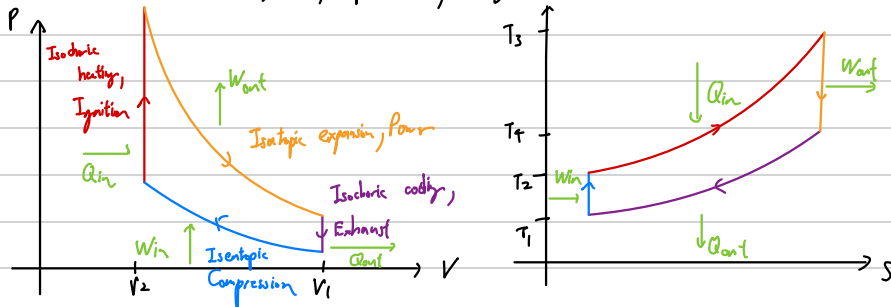
$$\frac{Q_h}{Q_c} \geq \frac{T_h}{T_c}$$

$$\text{Then } \text{COP} \leq \frac{T_c}{T_h - T_c}$$

4.3 Real Heat Engine

Otto Cycle (Internal combustion engine)

↳ No thermal reservoir, energy produced by burning fuel



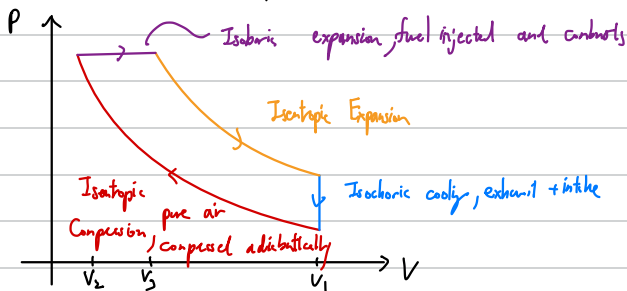
$$\hookrightarrow e = 1 - \left(\frac{V_2}{V_1} \right)^{\gamma-1} \quad \text{adiabatic exponent} \sim 30\% \text{ in real gasoline engines}$$

$$= 1 - \frac{T_1}{T_2} = 1 - \frac{T_4}{T_3}, \text{ since } TV^{\gamma-1} = \text{const}$$

Diesel Cycle (Engine)

↳ If fuel is too hot during compression, it might ignite itself before compression is complete

↳ Diesel engine start by compressing air then inject fuel into cylinder, hot air will combust the fuel w/o spark

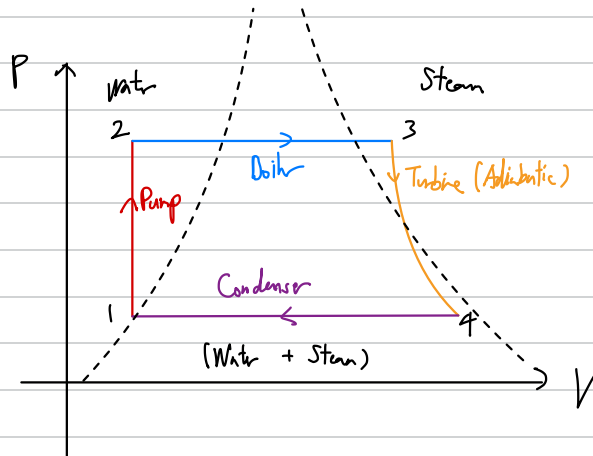
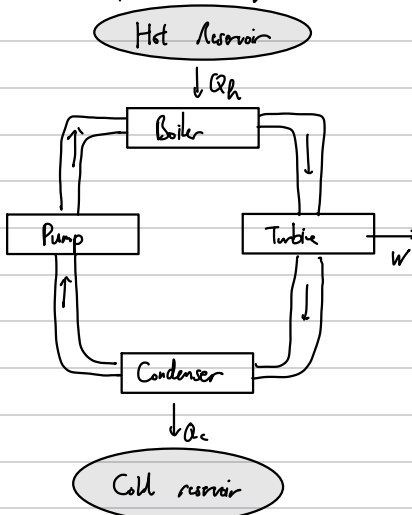


$$\hookrightarrow e = 1 - \left(\frac{V_2}{V_1} \right)^{\gamma-1} \left[\frac{\alpha^{\gamma}-1}{\gamma(\alpha-1)} \right]$$

$$\hookrightarrow \text{Cutoff Ratio, } \alpha = \frac{V_3}{V_2}$$

↳ Mathematically, the Otto cycle is more efficient, In practice the Diesel cycle is more efficient (up to 90%)

Rankine Cycle (Steam Engine)



$$\hookrightarrow W_{\text{net}} = (H_3 - H_4) - (H_2 - H_1)$$

$$Q_{\text{in}} = H_3 - H_2$$

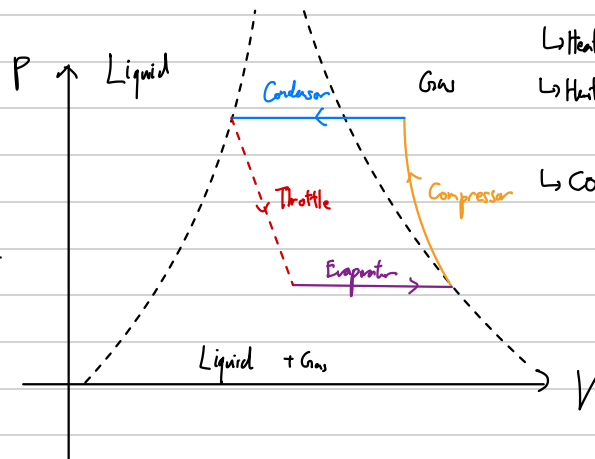
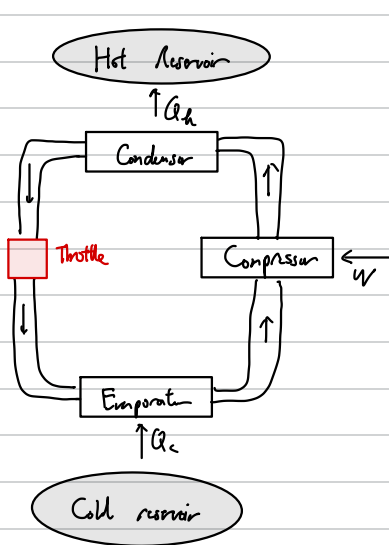
$$\hookrightarrow e = \frac{W_{\text{net}}}{Q_{\text{in}}} = 1 - \frac{H_4 - H_1}{H_3 - H_2} \approx 1 - \frac{H_4 - H_1}{H_3 - H_1}$$

↳ In Practice $H_2 \approx H_1$,

↳ small PV term for H in liquid.

4.4 Real Refrigerators

"Reversed" Rankine Cycle



↳ Heat absorbed in evaporator, $Q_{abs} = H_1 - H_4$

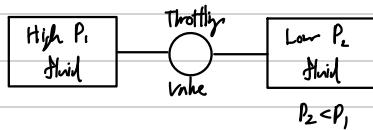
↳ Heat expelled in condenser, $Q_{exp} = H_2 - H_3$

$$\begin{aligned} \text{COP} &= \frac{Q_{abs}}{Q_{exp} - Q_{abs}} = \frac{H_1 - H_4}{H_2 - H_3 - (H_1 - H_4)} \\ &= \frac{H_1 - H_3}{H_2 - H_1} \end{aligned}$$

Throttling / Joule - Thomson Process

↳ When fluid at high pressure is forced through narrow restriction into lower pressure region.

↳ No heat added, no work done by fluid



↳ High Pressure $\rightarrow$ Low pressure, T drops
expand

↳ Energy exchange:

$$U_f - U_i = Q + W$$

$$= W_{P_1} + W_{P_2}$$

$$= P_1 V_1 - P_2 V_2$$

$$U_f + P_2 V_2 = U_i + P_1 V_1$$

$$H_f = H_i \quad \text{"isenthalpic"}$$

↳ For ideal gases, $\Delta H = 0$ implies $\Delta T = 0$, thus throttling only works on real fluids phase.

$$H = U_{potential} + U_{kinetic} + PV$$

for ex. Van der Waals force

Pressure $\downarrow \rightarrow$ Distance b/w molecules $\uparrow \rightarrow |U_{potential}| \downarrow \rightarrow$ Since $U_{pot} < 0$, $U_{kinetic} \downarrow \rightarrow T \downarrow$

$$\text{Then } \text{COP} = \frac{H_1 - H_3}{H_2 - H_1} \quad \text{for } H_3 = H_4$$

S.I Free Energy

Helmholtz free energy

$$F \equiv U - TS$$

energy needed to create the system minus the heat you can get for free from an environment at T

↳ The energy that must be provided as work if we create the system out of nothing ~~or~~
If we destroy the system, the maximum energy that you can retrieve.

Maximum Work

↳ For a reservoir at const temp T ,

$$dS_{\text{sys}} + dS_{\text{reservoir}} \geq 0 \quad \text{and}$$

$$dS_{\text{reservoir}} = -\frac{Q}{T}$$

$$\text{Then, } dS \geq \frac{Q}{T}, \quad Q \leq TdS$$

↳ From first law,

$$dU = Q + W$$

$$W \geq dU - TdS$$

↳ Helmholtz free energy, $dF = dU - TdS$

$$W \geq dF, \quad W_{\text{by system}} = -W$$

↳ Max work we can extract from a system at const T = drop in Helmholtz free energy

$$W_{\text{by}} \leq -\Delta F$$

Gibbs Free Energy

↳ For a system at const pressure, P ; temperature, T

$$G \equiv U - TS + PV = H - TS$$

building block Heat provided Energy needed to "make room"

↳ Relationship (Thermodynamic Potentials)

	$-TS$	
$+PV$	U	F
	H	G

↳ Work Relation,

$$\Delta G = \Delta U - T\Delta S + P\Delta V = Q + W - T\Delta S + P\Delta V$$

$$\left. \begin{array}{l} \textcircled{1} Q - T\Delta S \leq 0 \text{ always (second law)} \\ \textcircled{2} W \rightarrow \text{tot work} = \underbrace{-P\Delta V}_{\text{work done by environment}} + W_{\text{other}} \end{array} \right\} \longrightarrow \Delta G \leq W_{\text{other}}$$

Fuel Cells

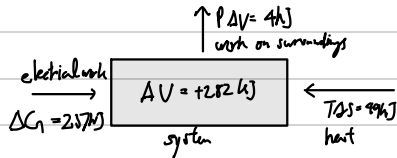
↳ Consider $\text{H}_2\text{O}_{\text{liquid}} \rightarrow \text{H}_{2\text{gas}} + \frac{1}{2}\text{O}_{2\text{gas}}$, $\Delta H = 286 \text{ kJ}$

↳ $P\Delta V = 4 \text{ kJ}$ is used to make room for the gas, the rest of 282 kJ remains in the system

↳ ΔS of system = $S_{\text{products}} - S_{\text{react}} = 165 \text{ J/K}$

Then the maximum amount of heat "siphoned" is $T\Delta S = 49 \text{ kJ}$ in room temp $\sim 298 \text{ K}$

↳ Thus ΔG (minimum amount of energy required to be supplied for the reaction to occur),
 $= \Delta H - T\Delta S = 237 \text{ kJ}$



↳ Reversing the process:

$\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$, extract 237 kJ of electrical work

$$e = \frac{286 - 49}{286} = 83\%$$

↳ Voltage = Work per electron

$$V = \frac{\Delta G}{n_e \cdot n_A}$$

5.2 Thermodynamic Identities, Gibbs energy as driving force

Identities

$$dF = -SdT - PdV + \mu dN \quad \text{and} \quad S = \left(\frac{\partial F}{\partial T} \right)_{V,N}, P = \left(\frac{\partial F}{\partial V} \right)_{T,N}, \mu = \left(\frac{\partial F}{\partial N} \right)_{T,V}$$

$$dG = -SdT - VdP + \mu dN \quad S = \left(\frac{\partial G}{\partial T} \right)_{P,N}, V = \left(\frac{\partial G}{\partial P} \right)_{T,N}, \mu = \left(\frac{\partial G}{\partial N} \right)_{T,P}$$

↳ Chemical potential:

For a system at const T, P

$$dG = \mu dN, \mu = G/N \quad (\text{Gibbs energy / particle})$$

Driving force

$$\text{Consider } \Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{sur}} \geq 0$$

$$\Delta S_{\text{sys}} - \Delta H/T \geq 0$$

$$\Delta H - T\Delta S \leq 0$$

$$\Delta G \leq 0$$

Reaction spontaneous for $\Delta G < 0 \rightarrow$ Entropy of universe increasing

$$\Delta G < 0 \rightarrow \Delta S_{\text{tot}} > 0 \quad \text{Spontaneous}$$

$$\Delta G = 0 \rightarrow \Delta S_{\text{tot}} = 0 \quad \text{Equilibrium}$$

$$\Delta G > 0 \rightarrow \Delta S_{\text{tot}} < 0 \quad \text{Impossible}$$

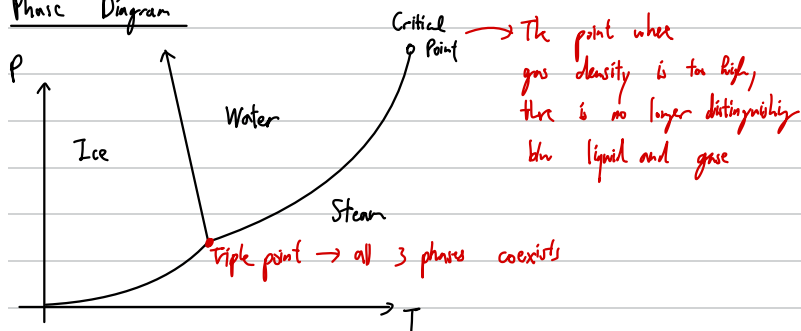
5.3 Phase Transformations of Pure Substances

Phase transformations

- ↳ Free energy used to drive liquid to gas transitions
- ↳ For const T, P the phase of lowest Gibbs free energy is stable

Entropy { Low T , system forms strong bonds (solid)
High T , system disordered (gases)

Phase Diagram

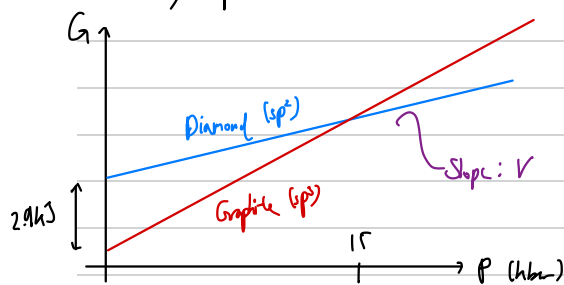


Type I Superconductors



- ↳ In superconducting phase, e^- from Cooper pairs $\rightarrow$ lowers G of the e^- system
- ↳ At low temperature, the ordered superconducting phase is preferred by the system over the disordered normal state
- ↳ Transition driven by minimizing thermodynamic potential, forming e^- pairs, not molecular bonds.

Diamonds, Graphite



- ↳ To turn from $sp^2 \rightarrow sp^3$, all covalent bond in the lattice simultaneously
- ↳ Thermodynamic shows direction
- ↳ Kinetics shows the time it takes (long, since massive kinetic barrier)

Temperature, Pressure dependence of Gibbs energy

↳ At const T, N ,

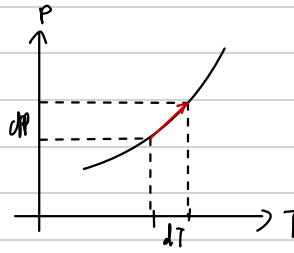
$$\left. \frac{\partial G}{\partial P} \right|_{T, N} = V \rightarrow \text{A substance with greater } V, \text{ greater } G \text{ as } P \uparrow$$

↳ At const P, N ,

$$\left. \frac{\partial G}{\partial T} \right|_{P, N} = -S \rightarrow \text{A substance with greater } S, \text{ smaller } G \text{ as } T \uparrow$$

Clausius - Clapeyron Relation

↳ At phase boundaries,
 $G_{\text{liquid}} = G_{\text{gas}}$ for equilibrium



↳ For increase by dT , dP , and phases to remain in equilibrium
 $dG_{\text{liquid}} = dG_{\text{gas}}$ for equilibrium

↳ Combine,

$$-S_{\text{liquid}} dT + V_{\text{liquid}} dP = -S_{\text{gas}} dT + V_{\text{gas}} dP \longrightarrow dG = dU - SdT + PdV$$

$$\frac{dP}{dT} = \frac{S_{\text{gas}} - S_{\text{liquid}}}{V_{\text{gas}} - V_{\text{liquid}}} = \frac{\Delta S}{\Delta V}$$

↳ In addition, $S = \frac{Q}{T}$, for $Q = L$, latent heat,

$$\boxed{\frac{dP}{dT} = \frac{L}{T\Delta V}} \quad \text{Clausius - Clapeyron relation}$$

↳ When ice $\rightarrow$ water $\Delta S > 0$, $\Delta V < 0 \rightarrow dP/dT = \Delta S/\Delta V < 0$, slope < 0

Ice - Skating

↳ Assuming a skate of 100kg $\sim$ 1000N

skate contact area $= 0.1\text{cm}^2 \sim 1 \times 10^{-7}\text{m}^2$

$$P \sim F/A = 10^8\text{Pa} \sim 1000\text{ bar}$$

↳ Melting temperature $\downarrow$ 7K at 1000 bar

↳ Ultimately, melting point of ice surface lower than bulk

↳ Skating is related to a different mechanism

5.4 Phase transformation of mixtures

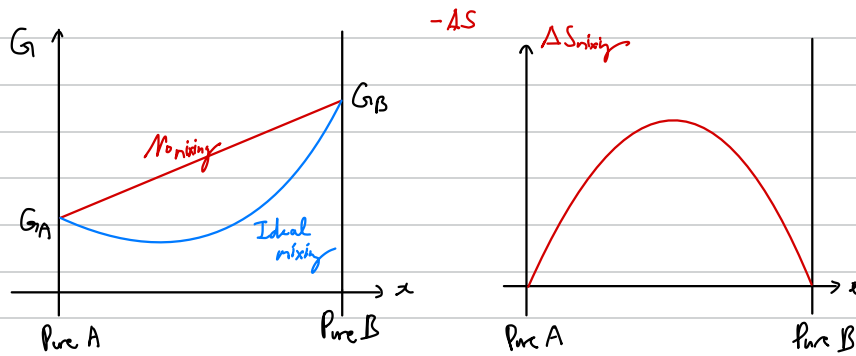
Before mixing

$$G = (1-x)G_A + xG_B \quad \xrightarrow{\text{proportion}} \quad x=0 \rightarrow \text{pure A}, x=1 \rightarrow \text{pure B}$$

Mixing

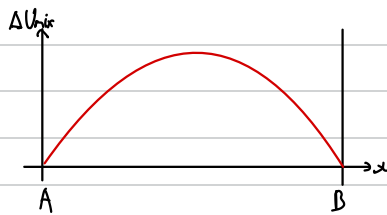
↳ Entropy increases as mixing is done $\rightarrow$ decrease in $G \rightarrow G$ of mixture $< G$ of pure components

$$G_{\text{mixture}} = (1-x)G_A + xG_B + T \left[x \ln x + (1-x) \ln (1-x) \right]$$



Non-Ideal Mixture

↳ Unlike molecules can be more less attracted to each other than like molecules (oil/water)



↳ For const P, V , $G = U + PV - TS$ only depend on $U, -TS$ tem.

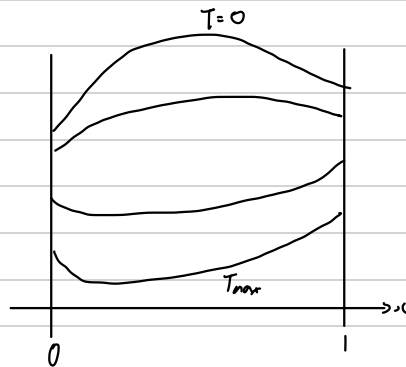
↳ $T=0$

At $T=0$, $\Delta G = \Delta U$

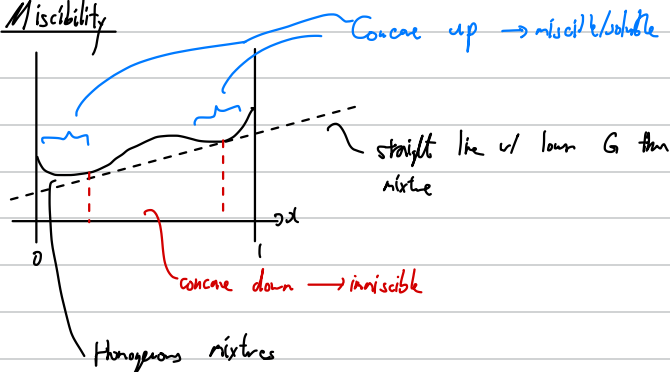
↳ $T>0$

Concave down contribution of ΔU competes with concave up of $-T\Delta S$

As $T \uparrow$, $-T\Delta S$ is always more significant.



Miscibility



5.5 Dilute Solutions

Dilute Solutions

- ↳ Primary component $\rightarrow$ solvent
- ↳ Other components $\rightarrow$ solutes
- ↳ Dilute when solute $\ll$ solvent

Chemical potentials and Free energy

- ↳ For pure A, Gibbs free energy:

$$G = N_A \mu_A(T, P) \quad (\text{pure solvent})$$

chemical potential, func. of T, P

- ↳ If we add a single B molecule,

$$dG = dU + PdV - TdS = f(T, P) - kT \ln N_A$$

terms independent on N_A

- ↳ For const P, T ,

$$dS = k \ln N_A + \text{terms independent of } N_A$$

direction of picking where to put the particle

- ↳ If we add 2 B molecule

$$dG \neq 2f(T, P) - 2kT \ln N_A \quad \text{since 2 B molecules are indistinguishable}$$

$$dG = 2f(T, P) - 2kT \ln N_A + kT \ln 2 \quad \text{divided multiply by 2}$$

- ↳ Adding N_B molecules,

account for indistinguishability

$$dG = N_B f(T, P) - N_B kT \ln N_A + kT \ln N_B! \approx N_B f(T, P) - N_B kT \ln N_A - kT N_B (\ln N_B - 1)$$

$$G = G_A + dG = N_A \mu_A(T, P) + N_B f(T, P) - N_B kT \ln N_A + N_B kT \ln N_B - N_B kT$$

★ Valid only for dilute solutions, since molecules interact w/ one another etc.

- ↳ For adding $N_C, N_D \dots$

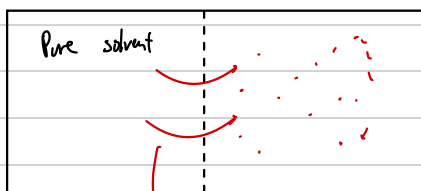
$$G = N_A \mu_A(T, P) + dG_B + dG_C + \dots$$

- ↳ Chemical potentials

$$\mu_A = \left(\frac{\partial G}{\partial N_A} \right)_{T, P, N_B} = \mu_A(T, P) - \frac{N_A kT}{N_A}$$

$$\mu_B = \left(\frac{\partial G}{\partial N_B} \right)_{T, P, N_A} = f(T, P) + kT \ln(N_B/N_A)$$

Osmotic Pressure



★ To prevent osmosis from happening, chemical potential left = right side

$$\mu_A(T, P_1) = \mu_A(T, P_2) - \frac{N_A kT}{N_A} \quad \text{and}$$

$$\mu_A(T, P_2) \approx \mu_A(T, P_1) + (P_2 - P_1) \frac{\partial \mu_A}{\partial P}$$

$$\text{Then, } (P_2 - P_1) \frac{\partial \mu_A}{\partial P} = \frac{N_A kT}{N_A}$$

★ van't Hoff's formula

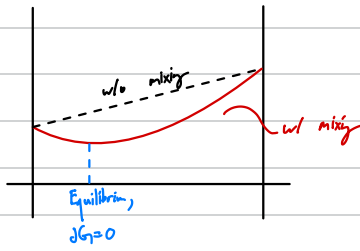
- ↳ For fixed T, N ,

$$\frac{\partial \mu_A}{\partial P} = \frac{\partial G}{\partial P} = \frac{V}{N} = \frac{V}{N_A}$$

$$(P_2 - P_1) = \frac{N_A kT}{V} = \frac{n_A RT}{V} \quad \text{moles of solute / volume}$$

5.6 Chemical Equilibrium

Water



↳ At equilibrium, $dG=0$, for fixed T, P

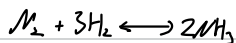
$$dG = \sum_i \mu_i dN_i = 0$$

↳ For each increase in OH^- , H^+ increases, H_2O decreases,

$$dN_{\text{H}_2\text{O}} = -1, dN_{\text{H}^+} = +1, dN_{\text{OH}^-} = +1$$

$$\text{Thus, } \mu_{\text{H}_2\text{O}} = \mu_{\text{H}^+} + \mu_{\text{OH}^-}$$

Ammonia



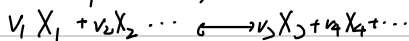
↳ At equilibrium, $dG=0$

$$dN_{\text{N}_2} = -1, dN_{\text{H}_2} = -3, dN_{\text{NH}_3} = +2$$

$$\mu_{\text{N}_2} + 3\mu_{\text{H}_2} = 2\mu_{\text{NH}_3}$$

Chemical equilibrium

↳ For any reaction,



↳ The equilibrium condition is

$$\nu_1 \mu_1 + \nu_2 \mu_2 + \cdots = \nu_3 \mu_3 + \nu_4 \mu_4 + \cdots$$

Nitrogen fixation

↳ For ideal gases,

$$\mu_X = \mu_X^\circ + kT \ln \left(\frac{P_X}{P^\circ} \right) \quad \text{usually later}$$

Standard chemical potential

$$\text{Then } \mu_{\text{N}_2} + kT \ln \left(\frac{P_{\text{N}_2}}{P^\circ} \right) + 3\mu_{\text{H}_2} + 3kT \ln \left(\frac{P_{\text{H}_2}}{P^\circ} \right) = 2\mu_{\text{NH}_3} + 2kT \ln \left(\frac{P_{\text{NH}_3}}{P^\circ} \right)$$

$$kT \ln \left(\frac{P_{\text{N}_2}}{P^\circ} \right) + 3kT \ln \left(\frac{P_{\text{H}_2}}{P^\circ} \right) - 2kT \ln \left(\frac{P_{\text{NH}_3}}{P^\circ} \right) = \underbrace{2\mu_{\text{NH}_3}^\circ - \mu_{\text{N}_2}^\circ - 3\mu_{\text{H}_2}^\circ}_{\text{standard Gibbs free energy, } \Delta G^\circ} \rightarrow \text{hypothetical change in } G$$

i.e. 1 mole of N_2 reacts with
 3 mole of H_2 to form
 2 mole of NH_3

$$\Delta G^\circ = RT \ln \left[\frac{P_{\text{N}_2} P_{\text{H}_2}^3}{P_{\text{NH}_3}^2 (P^\circ)^4} \right]$$

↳ Law of Mass action

$$\frac{P_{\text{NH}_3}^2 P^\circ}{P_{\text{N}_2} P_{\text{H}_2}^3} = e^{-\Delta G^\circ / RT} = K, \text{ equilibrium const. function of Temperature, } \Delta G^\circ(T) \text{ and } K$$

Dimensionless ratio of partial pressures

P° , reference pressure used to make K dimensionless

Le Chatelier's Principle

When a system in equilibrium is disturbed, it responds in a way that partially offsets the disturbance

ex. if $P \uparrow$, more H_2, N_2 forms NH_3 , decreasing particle #, $P \downarrow$

Water dissociation

↳ In equilibrium,

$$K = e^{(\Delta G^\circ / RT)} = 1 \times 10^{-14} \quad \text{and} \quad \Delta G^\circ = 74.9 \text{ kJ}$$

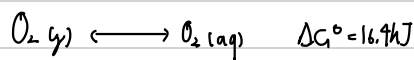
$$m_{H^+} \cdot m_{OH^-} = K, \quad m_{H^+} = m_{OH^-} \text{ in equilibrium}$$

molarities (mole solute per kg solvent)

$$\text{pH} \equiv -\log m_{H^+}$$

In equilibrium, $m_{H^+} = m_{OH^-} = 1 \times 10^{-7}$

Dissolving Oxygen in Water



↳ At equilibrium,

$$\mu_{\text{gas}} = \mu_{\text{aq}}$$

$$\mu_{\text{gas}}^\circ + RT \ln(P/P^\circ) = \mu_{\text{aq}}^\circ + RT \ln m$$

↳ Rearrange to give

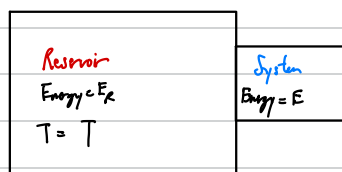
$$\frac{m}{P/P^\circ} = e^{-\Delta G^\circ / RT}, \quad \text{where } \Delta G^\circ = \mu_{\text{aq}}^\circ - \mu_{\text{gas}}^\circ$$

molar gas

Henry's Law $\rightarrow$ Ratio of dissolved O_2 to amount of adjacent gas is const. at any temperature or total pressure

6.1 Boltzmann Factor

Boltzmann Factor



↳ Hydrogen has: (negatively spin)

1 level at -13.6 eV

4 levels at -3.4 eV

9 levels at $-1.5 \text{ eV} \dots$

Degenerate levels (one energy level corresponds to multiple states)

↳ For an isolated system, all microstates are equally probable,
(particle + reservoir)

$\Omega_R(s_1) \rightarrow$ state of reservoir is s_1

↳ If s_1 has lower energy, $\Omega_R(s_1) > \Omega_R(s_2)$ to make up for it

↳ The ratio of probability is

$$\frac{P(s_2)}{P(s_1)} = \frac{\Omega_R(s_2)}{\Omega_R(s_1)}$$

$$\hookrightarrow S = k_B \ln \Omega \iff \Omega = e^{S/k_B}$$

Then,

$$\frac{P(s_2)}{P(s_1)} = e^{\frac{[S_R(s_2) - S_R(s_1)]}{k_B}} \text{ change in entropy of the reservoir (thy)}$$

$$\hookrightarrow \Delta S_R = \frac{1}{T} (A U_R + P \Delta V_R - \mu \Delta N_R)$$

① Multiply by dU_R to dU_R

② Zero for small systems

Thus,

$$S_R(s_2) - S_R(s_1) = \frac{1}{T} [U_R(s_2) - U_R(s_1)] = -\frac{1}{T} [E(s_2) - E(s_1)]$$

$$\hookrightarrow \text{Finally, } \frac{P(s_2)}{P(s_1)} = e^{-[E(s_2) - E(s_1)]/k_B T}$$

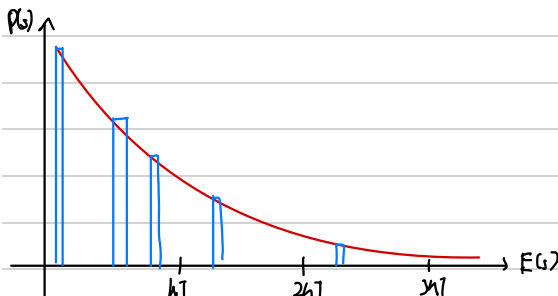
$$\text{Boltzmann factor} = e^{-E(s)/k_B T}$$

$$\hookrightarrow \text{Since } \frac{P(s_1)}{e^{-E(s_1)/k_B T}} = \frac{P(s_2)}{e^{-E(s_2)/k_B T}} = \frac{1}{Z}$$

Probability of each state s_i ,

$$P(s_i) = \frac{1}{Z} e^{-E(s_i)/k_B T}$$

Boltzmann Distribution



Partition Function, Z

$\hookrightarrow$ Total probability must be 1,

$$1 = \sum_s P(s) = \sum_s \frac{1}{Z} e^{-E(s)/k_B T}$$

$$Z = \sum_s e^{-E(s)/k_B T}, \text{ partition function}$$

$\hookrightarrow$ Usually just the first term, ignore higher terms

Z as "Counter"

$\hookrightarrow$ If $E_0 = 0$, the ground state has Boltzmann factor $e^0 = 1$, all excited states have factors < 1

$\hookrightarrow Z$ is the sum of all Boltzmann factors, counts how many available states

$\hookrightarrow$ Ground state = 1, factorial count for anything beyond

$\hookrightarrow$ At T=0, $Z=1$; As $T \uparrow$, $Z \uparrow$, Z tells us how many states the system is actually using at a particular temperature

Low T , Boltzmann factor ≈ 0 , $Z \approx 1$

High T , $k_B T \gg \Delta E$, Boltzmann factors ≈ 1 , $Z \approx$ total number of states

Abs vs Relative energy

↳ Shift all energies by E_0 s.t. $E_0' \rightarrow E_0 + E_0$

$$\textcircled{1} Z_{\text{new}} = Z \times e^{-E_0/kT}$$

$$\textcircled{2} P_i = \frac{e^{-E_i/kT}}{Z_{\text{new}}} = \frac{1}{Z} e^{-\frac{E_i - E_0}{kT}}$$

↳ Only relative energy matters

6.2 Average Value
Average Energy

$$\begin{aligned} \hookrightarrow \bar{E} &= \sum_j E_j P_j \\ &= \frac{1}{Z} \sum_j E_j e^{-\beta E_j}, \quad \beta = \frac{1}{kT} \end{aligned} \quad \hookrightarrow \bar{E} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln Z$$

↳ For a system of N particles,
 $U = N\bar{E}$

Rotational energy for Distinguishable diatomic molecules

↳ Quantized

$$E(j) = j(j+1)\epsilon, \quad j=0, 1, 2, \dots$$

$$\epsilon \propto (\text{moment of inertia})^{-1}$$

Q.M: $\epsilon = \hbar^2 / 8\pi^2 I$, $I = \mu r^2 = \frac{m_1 m_2}{m_1 + m_2} r^2$ ^{masses} dist b/w atoms (bond length)

↳ degeneracy, for level j the degenerate states number = $2j+1$

↳ Partition func of E_{rot}

$$Z_{\text{rot}} = \sum_j (2j+1) e^{-\frac{E(j)}{kT}} = \sum_j (2j+1) e^{-j(j+1)\epsilon/kT}$$

$$\approx \int_0^\infty (2j+1) e^{-\frac{j(j+1)\epsilon}{kT}} dj = \frac{kT}{\epsilon} \quad \text{for } kT \gg \epsilon$$

$$\hookrightarrow \text{Then } \bar{E}_{\text{rot}} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = \frac{1}{\beta} = kT$$

Identical atoms

↳ Rotating the molecule by 180° does not change spatial config $(A) \leftrightarrow (A') \xrightarrow{180^\circ} (A') \leftrightarrow (A)$

↳ Using $\frac{1}{2}$ to account for half of the available states

$$Z_{\text{rot}} = \frac{kT}{2}, \quad \text{but } \bar{E}_{\text{rot}} = kT$$

6.3 Equipartition Theorem

Proof of Equipartition Theorem

$$\hookrightarrow E = c q^2$$

where $c \rightarrow \text{const}$

$q \rightarrow \text{coord/momentum variable (continuous)}$

$\hookrightarrow$ Assume each state is discretely spaced by infinitesimally small Δq .

Then,

$$Z = \sum e^{-\beta E} = \sum e^{-\beta c q^2}$$

$$= \frac{1}{\Delta q} \sum e^{-\beta c q^2} \Delta q$$

$$= \frac{1}{\Delta q} \int e^{-\beta c q^2} dq$$

$$= \frac{1}{\Delta q} \sqrt{\frac{\pi}{\beta c}}$$

$\hookrightarrow$ For average energy,

$$\bar{E} = -\frac{\partial \ln Z}{\partial \beta} \quad \text{and}$$

$$\ln Z = \ln \frac{1}{\Delta q} \sqrt{\frac{\pi}{\beta c}} = -\frac{1}{2} \ln(\beta)$$

$$\text{Thus, } \bar{E} = \frac{1}{2\beta} = \frac{1}{2} kT$$

6.4 Partition Function and free energy relation

$\hookrightarrow$ Helmholtz free energy

$$F = -kT \ln Z$$

Proof 1

$\hookrightarrow$ Start with:

① Gibbs Entropy Formula

$$S = -k \sum P_i \ln(P_i)$$

② Boltzmann Probability

$$P_i = \frac{1}{Z} e^{-\beta E_i}$$

Proof 2

$\hookrightarrow$ Start w/ definition of Helmholtz free energy,

$$F = U - TS$$

$$\hookrightarrow \text{Then, } \left. \frac{\partial F}{\partial T} \right|_{V,N} = -S$$

$\hookrightarrow$ Combine:

$$S = -k \sum P_i \ln\left(\frac{1}{Z} e^{-\beta E_i}\right)$$

$$= \underbrace{\beta \sum P_i E_i}_U + k \ln Z \underbrace{\sum P_i}_1$$

$$S = U/T + k \ln Z$$

$$-kT \ln Z = U - TS = F$$

$$\text{And, } \left. \left(\frac{\partial F}{\partial T} \right) \right|_{V,N} = \frac{F - U}{T}$$

$\hookrightarrow$ Assume $F = -kT \ln Z$

$$\frac{\partial (-kT \ln Z)}{\partial T} = -k \ln Z - kT \frac{\partial}{\partial T} \ln Z$$

$$= -k \ln Z - \left(-\frac{1}{kT} \right) \frac{1}{Z} \frac{\partial Z}{\partial T} \quad \rightarrow -\bar{E} = -U$$

$$= -k \ln Z - \frac{U}{T}$$

$$= \frac{-kT \ln Z}{T} - \frac{U}{T} = \frac{F - U}{T}$$

Thermal properties and partition function

$$\hookrightarrow dF = dU - d(TS)$$

$$= dU - TdS - SdT$$

$$= TdS - PdV + \mu dN - TdS - SdT$$

$$= -SdT - PdV + \mu dN$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_{\mu, N} \quad P = -\left(\frac{\partial F}{\partial V}\right)_{T, \mu} \quad , \quad \mu = \left(\frac{\partial F}{\partial N}\right)_{T, V}$$

6.6 Composite Systems' Partition Functions

Non-interacting distinguishable particles

For 2 particles

$$\hookrightarrow E_{\text{tot}} = E_1 + E_2, \text{ and}$$

For N particles,

$$Z_{\text{tot}} = Z_1 Z_2 \dots Z_N$$

$$Z_{\text{total}} = \sum_j e^{-\beta[E_1(s_1) + E_2(s_2)]} = \sum_j e^{-\beta E_1} e^{-\beta E_2}$$

$$= \sum_{s_1} \sum_{s_2} e^{-\beta E_1(s_1)} e^{-\beta E_2(s_2)}$$

$$= \sum_{s_1} e^{-\beta E_1(s_1)} \cdot \sum_{s_2} e^{-\beta E_2(s_2)} = Z_1 Z_2$$

Non-interacting indistinguishable particles

$\hookrightarrow$ Putting particle 1 in state A, particle 2 in state B =
Putting particle 2 in state A, particle 1 in state B

$\hookrightarrow$ Use factor $1/N!$ to account for double counting

$$Z_{\text{tot}} = \frac{1}{N!} Z_1^N$$

$\star$ Note this ignores same states, which is not significant in low density ordinary gases.

6.4 Maxwell Speed Distribution

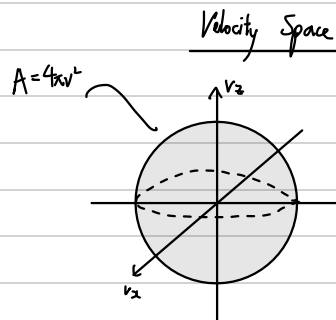
Velocity vs Speed

$\hookrightarrow$ For $\vec{v} = (v_x, v_y, v_z)$,

$$P(v_s) \propto e^{-E/kT}$$

$\hookrightarrow$ In addition, $E = \frac{1}{2}mv^2$, $v = |\vec{v}|$

$$P(v) \propto e^{-\frac{1}{2}mv^2/kT} \quad , \quad P(v)_{\text{max}} \text{ at } v=0$$



$\hookrightarrow$ Volume of states with speed v , $v \pm dv$ is
 $V = 4\pi v^2 dv$ $\star$ degeneracy factor

$\hookrightarrow P(v) dv = (\text{Boltzmann factor}) (\text{degeneracy factor})$

$\hookrightarrow$ But $\vec{v}_{\text{most probable}} = \vec{0}$ does not imply $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ is likely 0
velocity speed

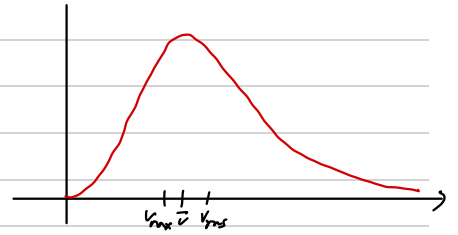
Characteristic Speeds

↳ Normal probability

$$\int_0^{\infty} A 4\pi v^2 e^{-mv^2/2kT} dv = 1 \longrightarrow A = \left(\frac{m}{2\pi kT}\right)^{3/2}$$

Normalization const

Distribution



↳ The velocity distribution:

$$P(v) = \left(\frac{m}{2\pi kT}\right)^{3/2} 4\pi v^2 e^{-mv^2/2kT}$$

Most Probable Speed

↳ For $dP/dv = 0$,

$$v_p = \sqrt{\frac{2kT}{m}}$$

Average Speed

$$\bar{v} = \int_0^{\infty} v P(v) dv = \sqrt{\frac{8kT}{\pi m}}$$

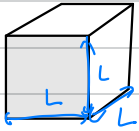
RMS speed

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{2L}$$

6.7 Ideal Gas Revisited

Single atom partition (1D)



↳ Molecule wavefunction wavelengths,

$$\lambda_n = \frac{2L}{n}, n=1,2,\dots$$

$$E_n = \frac{p_n^2}{2m} = \frac{h^2 n^2}{8mL^2}$$

$$Z = \sum_n e^{-E_n/kT} = \sum_n e^{-h^2 n^2 / 8mL^2 kT}$$

$$\approx \int_0^{\infty} e^{-h^2 n^2 / 8mL^2 kT} dn = \sqrt{\frac{2\pi mkT}{h^2}} L = \frac{L}{\lambda_Q}$$

3D Case (1 atom)

$$E = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)$$

$$Z = \int \int \int_{V^3} e^{-\beta \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)} dV$$

$$= \left[\frac{L}{\lambda_Q} \right]^3$$

↳ Quantum volume, $v_Q = \lambda_Q^3$

↳ Threshold where classical physics break down

↳ Quantum volume of a cube whose length = λ_Q i.e. $E = kT$

↳ $Z_1 = \frac{V}{v_Q}$, number of de Broglie wavelength cubes can fit inside volume of the box
Translational Partition function

↳ Thermal De Broglie Wavelength, $\lambda_Q = \frac{h}{\sqrt{2\pi mkT}}$

Molecules

$$Z_N = \frac{1}{N!} (z_1)^N = \frac{1}{N!} \left(\frac{V}{v_Q} \right)^N$$

N molecules

$$Z_N = \frac{1}{N!} \left(\frac{V}{v_Q} z_{int} \right)^N$$

account for rotation, vibration etc

↳ holds if $\frac{V}{N} \gg v_Q$ (dilute gas)

↳ clac cor QM (Doe Fermi)

Thermodynamic relations

$$U = -\frac{\partial}{\partial \beta} \ln Z$$

$$C_V = \frac{\partial U}{\partial T} = \frac{\partial U_{int}}{\partial T} + \frac{5}{2} Nk$$

$$P = -\frac{\partial F}{\partial V} = \frac{NkT}{V}$$

$$= -NkT \ln Z_{int} + NkT \frac{\partial \ln Z_{int}}{\partial T}$$

$$F = -kT \ln Z$$

$$S = -\left(\frac{\partial F}{\partial T} \right)_{V,N} = Nk \left[\ln \left(\frac{V}{N v_Q} \right) + \frac{5}{2} \right] - \frac{\partial F_{int}}{\partial T}$$

$$= N E_{int} + \frac{3}{2} NkT$$

internal energy *avg trans. energy*

$$= NkT [4V - \ln N - \ln v_Q + 1] + F_{int}$$

$F_{int} = -NkT \ln Z_{int}$, internal degrees of freedom contribution

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T,V} = -kT \ln \left(\frac{V}{N v_Q} \right)$$

Derivation

Ideal Gas Chemical Potential

$$G = \sum N_i \mu_i$$

$$\frac{d\mu}{dP} = \frac{d\mu}{dG} \cdot \frac{dG}{dP} = \frac{V}{N}$$

$$d\mu = \frac{kT}{P} dP$$

$$\mu(T, P) - \mu(T, P^0) = kT \ln(P/P^0)$$

$$\mu(T, P) = \mu^0 + kT \ln(P/P^0)$$

Clausius-Clapeyron Relation

At boundaries, $dG_g = dG_l$

$$-S_g dT + V_g dP = -S_l dT + V_l dP$$

$$(V_g - V_l) dP = (S_g - S_l) dT \longrightarrow L = T \cdot S$$

$$\frac{dP}{dT} = \frac{1}{V_g - V_l} \cdot \frac{L}{T}$$

$$\frac{dP}{dT} = \frac{L}{T \Delta V}$$

Entropy of Mixing

$$\Delta S = k \ln \left(\frac{N}{N_A} \right)$$

$$= k \ln \left[N! / (N_A! N_B!) \right]$$

$$= k \ln [x \ln x + (1-x) \ln(1-x)]$$

Chemical Potential

Pure

$$\text{Solvent: } \mu_0(T, P) - \frac{N_A k T}{N_A}$$

$$\text{Solute: } \mu^0(T, P) + kT \ln m_B$$

Osmotic Pressure van't Hoff's eq

$$\mu_{\text{pure solvent}} = \mu_{\text{solute}}$$

$$\mu_0(T, P_1) = \mu_0(T, P_2) - \frac{N_A k T}{N_A}$$

$$\mu_0(T, P_1) = \mu_0(T, P_1) + \frac{\partial \mu}{\partial P} (P_2 - P_1) - \frac{N_A k T}{N_A}$$

$$\text{Since } N_A \gg N_B, N \approx N_A$$

$$\text{and } \frac{\partial \mu}{\partial P} = \frac{\partial \mu}{\partial G} \cdot \frac{dG}{dP} = \frac{1}{N} \cdot V$$

$$\frac{V}{N_A} (P_2 - P_1) = \frac{N_A k T}{N_A}$$

$$(P_2 - P_1) = \frac{N_A k T}{V}$$

Boiling/Freezing Point

Raoult's Law

$$\mu_{A, \text{liq}}(T, P) = \mu_{A, \text{gas}}(T, P)$$

$$\cancel{\mu_{A, \text{liq}}(T, P_0)} + \frac{\cancel{d\mu_{A, \text{liq}}(P - P_0)}}{dP} - \frac{kT N_A}{N_A} = \cancel{\mu_{A, \text{gas}}(T, P_0)} + \frac{\cancel{d\mu_{A, \text{gas}}(P - P_0)}}{dP}$$

$$\cancel{\mu_{A, \text{liq}}(T_0, P)} + \frac{\cancel{d\mu_{A, \text{liq}}(T - T_0)}}{dT} - \frac{kT N_A}{N_A} = \cancel{\mu_{A, \text{gas}}(T_0, P)} + \frac{\cancel{d\mu_{A, \text{gas}}(T - T_0)}}{dT}$$

$$\text{Since } \frac{d\mu}{dP} = \frac{V}{N}$$

$$\left(\frac{V}{N} \right)_l (P - P_0) - \frac{kT N_A}{N_A} = \left(\frac{V}{N} \right)_g (P - P_0)$$

$$- \frac{kT N_A}{N_A} = \left(\frac{V}{P_0} \right) (P - P_0)$$

$$- P_0 \frac{N_A}{N_A} = P - P_0$$

$$\frac{P}{P_0} = 1 - \frac{N_A}{N_A}$$

$$\text{Since } \frac{d\mu}{dT} = \frac{d\mu}{dG} \cdot \frac{dG}{dT} = -\frac{S}{N}$$

$$\left(-\frac{S}{N} \right)_l (T - T_0) - \frac{kT N_A}{N_A} = \left(-\frac{S}{N} \right)_g (T - T_0)$$

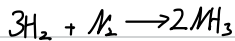
$$\text{And } S_l - S_g = \frac{L}{T_0}, N \sim N_A$$

$$\left(\frac{L}{T_0 N_A} \right) (T - T_0) = \frac{kT N_A}{N_A}$$

$$T - T_0 = \frac{kT_0^2 N_A}{L} \quad T_0 \approx T$$

Equilibrium constant

Ammonia:



At equilibrium,

$$3\mu_{\text{H}_2} + \mu_{\text{N}_2} = 2\mu_{\text{NH}_3}$$

$$3[\mu_{\text{H}_2} + kT \ln(P_{\text{H}_2}/P_0)] + [\mu_{\text{N}_2} + kT \ln(P_{\text{N}_2}/P_0)] = 2[\mu_{\text{NH}_3} + kT \ln(P_{\text{NH}_3}/P_0)]$$

$$3\mu_{\text{H}_2} + \mu_{\text{N}_2} - 2\mu_{\text{NH}_3} = kT \ln \left[\frac{P_{\text{NH}_3}^2 P_0^2}{P_{\text{H}_2}^3 P_{\text{N}_2} P_0^2} \right]$$

$$-\Delta G^0 = RT \ln \left[\frac{P_{\text{NH}_3}^2 P_0^2}{P_{\text{H}_2}^3 P_{\text{N}_2} P_0^2} \right]$$

since G
= prod - react

$$e^{-\Delta G^0/RT} = K$$

Dissociation

$$\mu_{\text{H}_2\text{O}} = \mu_{\text{H}^+} + \mu_{\text{OH}^-}$$

$$\mu_{\text{H}_2\text{O}}^0 = \mu_{\text{H}^+}^0 + kT \ln(m_{\text{H}^+}) + \mu_{\text{OH}^-}^0 + kT \ln(m_{\text{OH}^-})$$

$$\mu_{\text{H}_2\text{O}}^0 - \mu_{\text{H}^+}^0 - \mu_{\text{OH}^-}^0 = kT \ln(m_{\text{H}^+} m_{\text{OH}^-})$$

$\Delta \mu$ for solute

$$e^{-\Delta G^0/RT} = K$$

Dissolving

$$\mu_{\text{gas}} = \mu_{\text{solute}}$$

$$\mu_{\text{gas}} + kT \ln(P/P_0) = \mu_{\text{solute}}^0 + kT \ln(m)$$

$$RT \ln \left(\frac{P/P_0}{m} \right) = \Delta G^0$$

$$\frac{m}{P/P_0} = e^{-\Delta G^0/RT}$$